SAPS

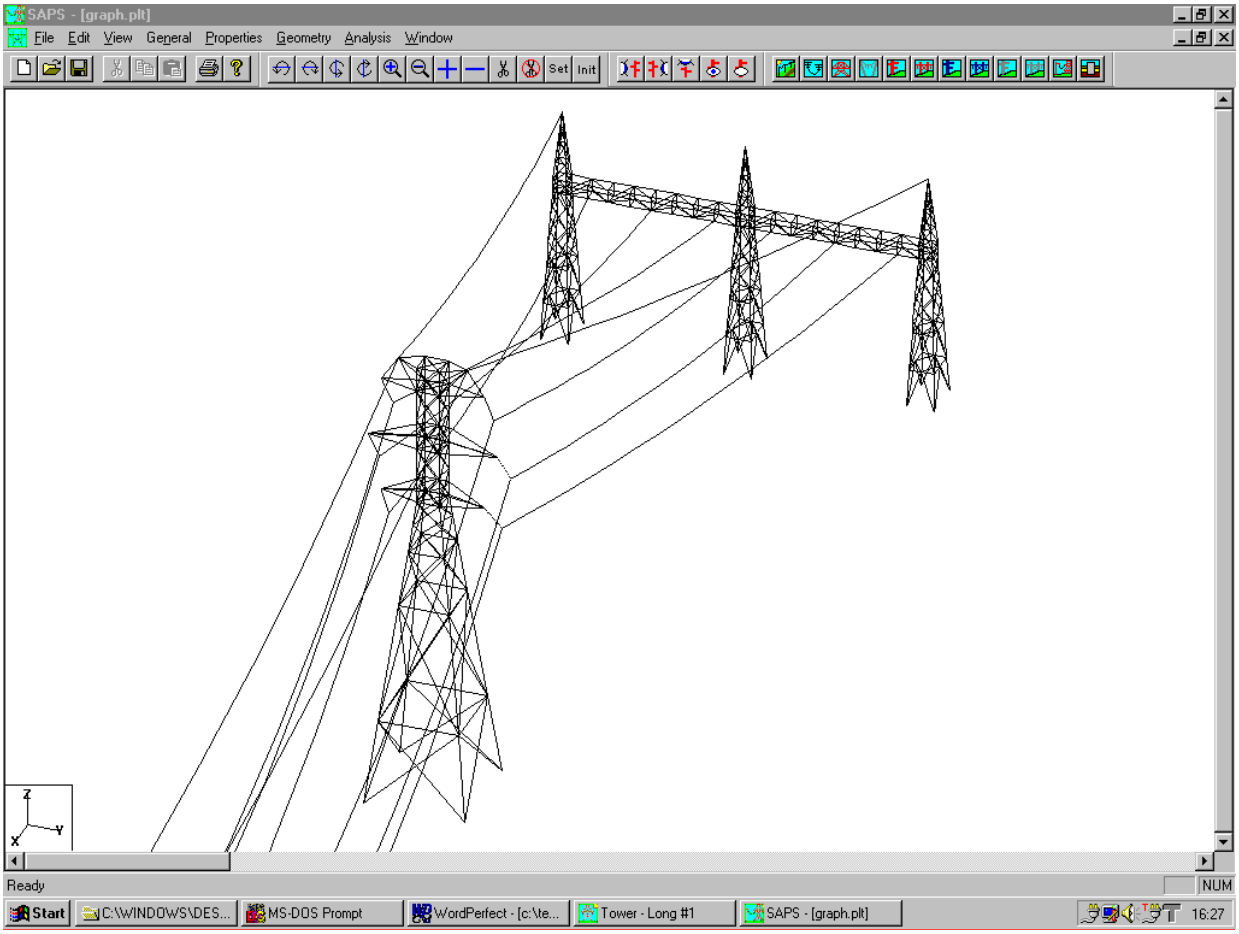

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TABLE OF CONTENTS

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DISCLAIMER, WARRANTY AND LICENSES 6

1. OVERVIEW 7

1.1 Typical applications 7
1.2 Features summary 8
1.3 Online help and electronic version of manual 10
1.4 Customizing menus, dialog boxes, tables and reports 10
1.5 Attachments and annotations 10

2. MODELING AND ANALYSIS ASSUMPTIONS 11

2.1 Coordinate systems 11
2.2 Joints 12
2.2.1 Joint degrees of freedom 12
2.2.2 Tied joints 13
2.2.3 Master-slave joints 13
2.2.4 Differences between tied and slave joints 14
2.3 Elements or members 15
2.3.1 Truss element 15
2.3.2 Beam element 15
2.3.3 Cable element 18
2.3.4 Fuse element 18
2.3.5 Linear substructure 19
2.4 Loads 21
2.4.1 Dead, temperature and ice loads 21
2.4.2 Wind (or current) loads 21
2.4.2.1 Special override for cable elements 22
2.4.3 Total distributed load on element 23
2.4.4 Concentrated forces or moments 23
2.4.5 Specified displacements or rotations at supports 23
2.5 Solution strategies for nonlinear problems 24
2.5.1 Basic Newton-Raphson method 24
2.5.2 Modified Newton-Raphson method with stiffness addition 25
2.5.3 Newton-Raphson in combination with load increment 26
2.5.4 Ill-conditioned stiffness matrix 26
2.6 Strength checks 28

2.6.1 Truss, cable and fuse elements 28
2.6.2 Beam elements 28
2.6.2.1 Axial check only 28
2.6.2.2 Stress check #1 28
2.6.2.3 Stress check #2 29
2.6.2.4 Interaction check #1 29
2.6.2.5 Interaction check #2 29

3. LOADING, RUNNING AND VIEWING EXISTING MODEL 30

3.1 The File menu 30
3.1.1 Directory preferences and units 31
3.1.2 Saving or backing-up model 33
3.2 Model description 34
3.2.1 Viewing model in Initial Geometry window 34
3.2.2 Model creation and editing 37
3.3 Running the analysis 37
3.4 Analysis report 37
3.4.1 Viewing, printing and exporting 38
3.4.2 Customizing 38
3.5 Deformed geometry window 39

4. BUILDING A MODEL 42
4.1 General menu 42
4.1.1 General data 42
4.1.2 Nonlinear convergence parameters 43
4.2 Element properties data bases 44
4.2.1 Truss/ Fuse elements 44
4.2.2 Beam elements 45
4.2.3 Cable elements 46
4.2.4 Substructures 46
4.3 Geometry 47
4.3.1 Joints 47
4.3.2 Elements connectivity 48
4.3.2.1 Truss/ Fuse elements 48
4.3.2.2 Beam elements 49
4.3.2.3 Cable elements 50
4.3.2.4 Substructures 50
4.4 Load cases 52
4.4.1 Concentrated loads 53
4.4.2 Specified support displacements 53
4.4.3 Element temperatures 54

4.4.4 Wind description 55
4.4.5 Ice loads 59
4.4.6 Load options 60

5. EXAMPLES 61

5.1 Example 1 - Guyed latticed tower 62
5.2 Example 2 - Steel frame 63
5.3 Example 3 - Flexible steel frame line segment (detailed modeling) 64
5.4 Example 4 - Flexible steel frame line segment (use of substructures) 65
5.5 Example 5 - Substructure properties generation 66
5.6 Example 6 - Line segment with different support conditions 67
5.7 Example 7 - Portion of latticed tower 68
5.8 Example 8 - Large antenna 69
5.9 Example 9 - Merging models 70

APPENDIX A HARDWARE REQUIREMENTS AND INSTALLATION 71

APPENDIX B REFERENCES 75

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1. OVERVIEW

SAPS is one of several **MS-WINDOWS** programs developed by Power Line Systems, Inc. for the analysis and design of transmission structures, systems or sub-systems. Others include **PLS-POLE**, **SAGSEC**, and **TOWER**. These programs have many features in common, including the use of similar input menus, reports and graphic displays. They all have linear or nonlinear structural analysis capabilities.

SAPS was first introduced in 1985 as a **MS-DOS** program and it has been improved ever since. Its algorithms are used as the analysis engines of our **PLS-POLE**, **SAGSEC**, and **TOWER** transmission structure programs. **SAPS** is also used internally by our line design program **PLS-CADD** whenever one requires a finite element analysis of the wire system in that program. A finite element analysis is more accurate than a "ruling span" analysis, as explained in the **PLS-CADD** manual.

The name **SAPS** was chosen as the acronym for **S**tructural **A**nalysis of **P**ower and communication **S**ystems. Even though **SAPS** is an absolutely general structural analysis program, its primary intended use is in the area of transmission and communication systems. Transmission and communication systems generally include cables in the form of conductors, ground wires, guys, suspension strings, etc. They often include flexible components which deflect significantly under working conditions. The presence of cables and large deflections requires a geometric nonlinear analysis.

SAPS has been used by many fabricators, utilities and consultants throughout the world. A program similar to **SAPS**, called **SOUPLE** (Peyrot, 1988), was developed for offshore applications.

Our specialized transmission structure programs (**PLS-POLE**, **SAGSEC**, and **TOWER**) and our **PLS-CADD** program automatically develop the mathematical model (joints and members) of your structure or wire system from overall geometric and component properties. However, with **SAPS**, you need to build the mathematical model yourself. This requires more effort but provides you with much more flexibility to handle complex structures. In general you are much better off using **PLS-CADD**, **PLS-POLE** and **TOWER** than **SAPS**.

1.1 Typical applications

Applications of **SAPS** are almost limitless. **SAPS** can be used for off-shore as well as land-based systems. In off-shore applications, member forces produced by moving water are automatically calculated from values of its mass density and velocity field. In land-based applications, the member forces are generated from wind and/or ice accretion.

Examples of transmission or communication systems which have been analyzed with **SAPS** include:

* Lattice towers (with or without "tension-only" assumption)

* Guyed Delta, Y, or V structures (ASCE, 1997)
* Cross rope suspension structures (Peyrot, 1981)
* Sags, tensions and loads in multi-spans systems
* Flexible concrete, steel or wood poles
* Multi-pole structures (H-Frame, 3-Pole guyed corner structure, stub poles, etc.)
* Tall guyed antenna masts (Example 8)
* Guyed microwave towers with or without stabilizers
* Antenna arrays or walls
* Suspension bridges
* Substation structures
* Effects of unbalanced ice, broken conductor, broken guy, or broken insulator (Peyrot, 1980)
* Effects of slack changes due to offset clipping, splicing, etc., in line segments
* Complete line segments
* Flexible poles or frames interconnected by cables (Peyrot, 1980; Mozer, 1983)

Examples of off-shore systems include:

* Mixed mooring lines with cables, chains, buoys and hangers (Peyrot, 1980)
* Towed cable arrays
* Towed underwater buggies or sonars
* Ocean Thermal Energy Conversion (OTEC) systems
* Tension leg platforms or underwater guyed towers
* Production risers
* Floating platforms
* Flexible pipe systems

1.2 Features summary

The key features that make **SAPS** such a powerful and unique program are:

* Linear or non-linear analysis capability
* Truss, beam, cable, fuse elements
* Substructures (super-elements)
* Automatic substructure creation
* Kinematic coupling of joints
* Master-slave joint relationships
* Automatic bandwidth minimization
* Automatic calculation of wind, ice and thermal loads on members
* Adjustable wind or water velocity profiles
* Support settlements
* Powerful graphics capability
* Control of convergence process (iteration strategy)

* Component databases
* Interactive model building
* Ability to copy/paste entire structure to build complete line model

All input data for **SAPS** are entered through interactive menus and tables. These data describe the geometry of the model, the initial sagging condition of the cables and the loading assumptions. The model initial geometry is shown graphically in the **Initial Geometry** window prior to analysis. Once the analysis is completed, the results are available in text reports and in the **Deformed Geometry** window. Fig. 1-1 shows the **Deformed Geometry** window for the last span of a transmission line subjected to wind near a substation structure. You will note that the insulator swings are shown at scale and are different for different phases. Clearances from the conductor to the body of the tower can be measured graphically.

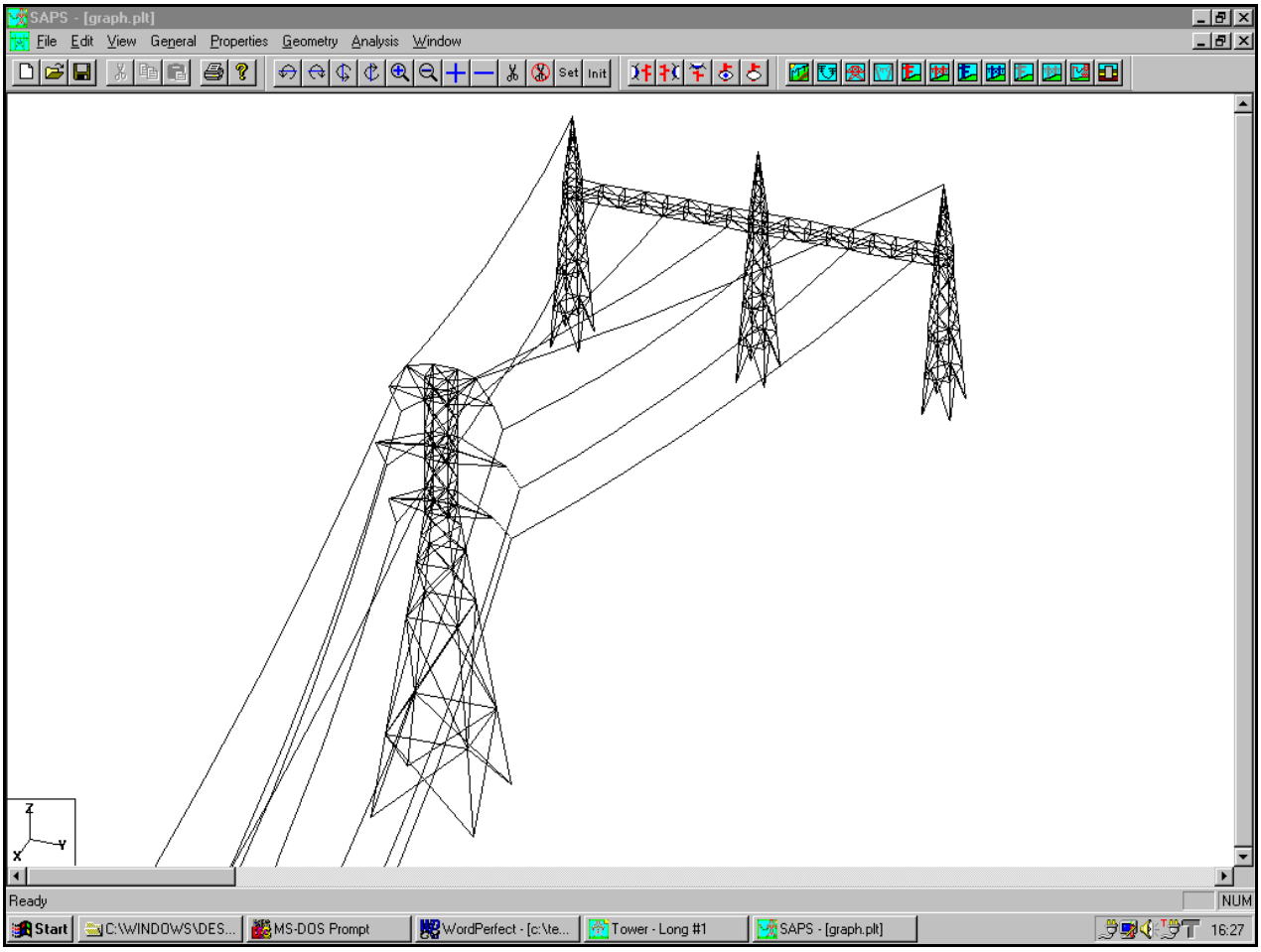


Fig. 1-1 Terminal span under wind load

Section 2 reviews some of the classical finite element modeling concepts which you need to be familiar with to use **SAPS**.

Section 3 shows you how to navigate through the menus.

Section 4 shows you how to build a model and Section 5 shows some examples to illustrate the program capabilities.

Finally, Appendix A discusses hardware requirements and installation, and Appendix B lists the references.

1.3 Online help and electronic version of manual

You can download the latest electronic version of the manual (if you are within your free upgrade period) and view that manual by going to the Help menus.

Online help is available in any menu or dialog box. All you need to do is to click on the "?" button at the top right of the box and you will be taken to the appropriate section of this manual where you will find the appropriate information.

1.4 Customizing menus, dialog boxes, tables and reports

SAPS lets you customize menus, dialog boxes, tips, tables and reports, thus giving you complete flexibility to translate their text strings into the language or format of your choice. Since customization features are common to all our programs, we refer you to Appendix G of the **TOWER** manual for a full discussion.

1.5 Attachments and annotations

You can attach drawings described by DXF files and photographs to the view of your structure in **SAPS**. You also have limited line and text annotation capabilities. Since these feature are also shared with our other structure programs, we refer you to Section 4.2.7 of the **TOWER** manual for a full discussion on attachments and to Section 4.2.8 of that manual for a discussion on annotations.

2. MODELING AND ANALYSIS ASSUMPTIONS

The analysis performed internally by **SAPS** is based on classical concepts of structural analysis by finite elements. These concepts, which you should be familiar with, are described in most textbooks dealing the computerized structural analysis (Au, 1993; Bathe, 1982; Cook, 1981; Hoit, 1995; Wang, 1983)

The mathematical model which you build in **SAPS** is a collection of joints (or nodes) interconnected by members (or elements). The basic geometric dimensions are established by placing the joints on the structure. Each joint is given a unique identification label and is located in space with coordinates that are associated with a global 3-dimensional coordinate system. The model geometry is completed by connecting the joints with structural members. In **SAPS**, the member types are: truss, beam, cable, fuse and super-element (or substructure).

Some basic rules for the selection of the minimum number of joints and members for accuracy are:

* There should be enough joints to describe the geometry of the system
* Joints need to be located at points and lines of discontinuity, i.e. at changes in material properties, section properties, etc.
* Joints should be located at points on the structure where displacements are to be evaluated
* Joints should be located at points where concentrated loads are to be applied
* There should be a joint at each support location
* Wind and ice loads on truss, fuse and beam members are simulated automatically by equivalent concentrated forces applied at both ends of the members. Consequently, it is sometimes necessary to break a straight structural member into several elements (i.e. several intermediate joints) to better simulate the variation of the load along the member.

2.1 Coordinate systems

For the generation of all input data and the interpretation of the analysis results, it is necessary to deal with several reference axes. A 3-dimensional right-handed global (X, Y, Z) Cartesian coordinate system is used to specify the locations of the joints. Because **SAPS** calculates gravity loads automatically, it is necessary to always orient the <u>global Z-axis vertically up</u>, i.e. in the direction opposite to that of the pull of gravity. The axes X and Y are therefore always in an horizontal plane. Also, if a wind profile is specified, it is <u>imperative that the “Ground Elevation”, i.e. Z of the ground,</u> be entered in the **Wind Profile** dialog box of Fig. 4.14-a. This will allow the wind load on a member to be generated automatically, based on the average height of that member above the ground.

For input and output associated with beam members, a local 3-dimensional right handed (x, y, z) Cartesian system is used as described in Section 2.3.2. The local coordinate system is tied to the beam member. The beam properties, as well as the output moments, shears and stresses are given

with reference to the local axes.

2.2 Joints

2.2.1 Joint degrees of freedom

Every joint in a three-dimensional structure has a maximum of three possible displacements and three possible rotations. Namely, the joint can translate in the global X-, Y-, and Z-directions and it can also rotate about the X-, Y-, and Z-axes. If, because of loads on the structure, a joint moves or rotates in or about any of the three directions, the joint is said to have a Degree Of Freedom (DOF) in translation or rotation in that direction. Rotations are just generalized cases of displacements and will be referred to as displacements from here on.

If the displacement (or rotation) of a joint in any of the three X-, Y-, Z-directions is known to be zero, such as at a support point, then there is no DOF in that direction. The joint is fixed in that direction. Therefore, a joint may have a total of from zero to six DOF's.

Since the forces in truss, cable and fuse elements depend only on the location of their end joints (and not on the joint rotations), it is not necessary to consider rotational DOF's at joints where only truss, cable or fuse elements meet. For example, the three arrows in Fig. 2-1.a show the three translational DOF's which are sufficient to describe the position of the joint at the junction of cable elements A and B.

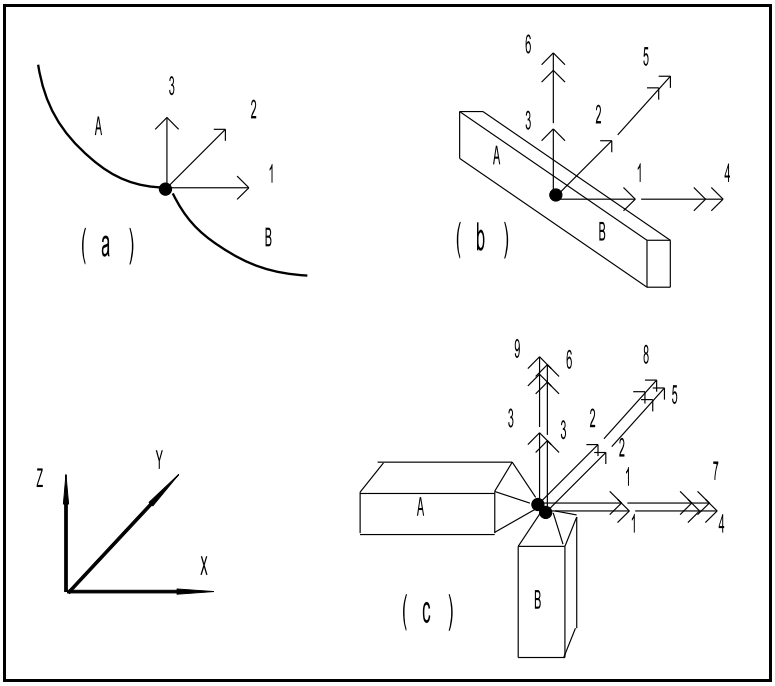


Fig. 2-1 Joint degrees of freedom

Since the forces and moments at the ends of a beam element depend on both the location and rotation of its end joints, translational and rotational DOF's should be considered at a joint where one or more beam elements terminate (Fig . 2-1.b).

Hinges behaving as universal joints should be modeled with a pair of joints at the same location. For example, the hinge at the junction of beam elements A and B in Fig. 2-1.c is modeled with one joint (master joint) at the end of element B having three DOF's numbered 1 to 3 and three rotational DOF's numbered 5 to 6 plus another joint (tied joint) at the end of element A sharing the same DOF's 1 to 3

with the master joint but having three independent rotational DOF's numbered 7 to 9. Therefore, a total of nine DOF's are needed to model a universal joint. A tied joint shares some of its DOF's of freedom with its master joint, as discussed further in Section 2.2.2.

The mathematical model developed internally by **SAPS** assigns one specific DOF number to each joint in the structure. The total number of DOF's is also the number of separate simultaneous equations that the program has to solve at each step of the solution process. **SAPS** uses a band solver for the solution of the simultaneous equations and it optimizes the numbering of the DOF's for efficient computer memory usage and solution time.

2.2.2 Tied joints

SAPS has the ability to tie (or couple) joint DOF's together. The need for two joints to model a hinge (universal joint) was discussed in Section 2.2.1. Consider now the example of a bolted connection between a vertical pole and an horizontal arm as shown in Fig. 2-2. The connection can be modeled by two joints. The first joint, arbitrarily selected as the master joint, is at the intersection of the two vertical beam elements representing segments of the pole above and below the connection. The second joint, or tied joint, is at the intersection of the two horizontal beam elements to the left and to the right of the connection. If the pole and arm were rigidly connected, both joints would be constrained to translate by the same amount and also to rotate about the X-, Y- and Z-axes by the same amount. In that fixed situation, the second joint would be a slave to the first joint as described in Section 2.2.3. However, with a single bolt connection, relative rotations between the pole and arm are possible around the bolt. Therefore, the rotations of the master and tied joints about the X-axis may be different. This can be modeled in **SAPS** by constraining the tied joint to have the same degrees of freedom as the master joint, except for the tied joint rotational degree of freedom about the X-axis which is independent of the corresponding master joint DOF.

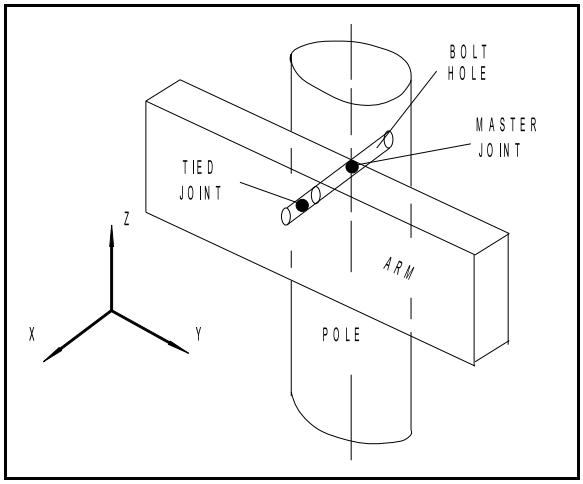


Fig. 2-2 Tied joints

2.2.3 Master-slave joints

SAPS also implements the concept of master-slave relationships between joints. A joint is said to be the slave of a master joint if its position is entirely defined by the position of the master joint. This occurs whenever it can be assumed that the material between a slave and its master is infinitely rigid. A slave joint has no degree of freedom, since its motion is entirely controlled by the master. One master may have several slaves, but one slave is allowed only one master.

For example, joint A in Fig. 2-3.b is the attachment point of a guy to the face of the pole. Joint C is the joint at the same level on the pole centerline. Vertical beam members making up the pole connect at joint C. Assuming no deformation in the pole material between joints C and A, the movement of A is entirely determined by that of C. Therefore A is slave to C.

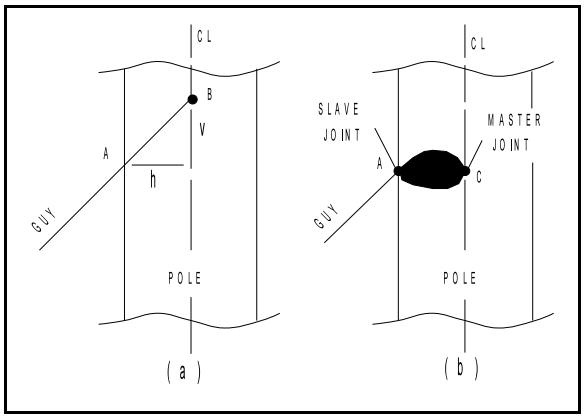


Fig. 2-3 Master-slave joints

2.2.4 Differences between tied and slave joints

There are fundamental differences between tied joints and slave joints, even though each one has a master.

If joint A in Fig. 2-3.b is modeled as a slave to C, a translational displacement at C produces an identical displacement at A, but a rotation at C causes a rigid body rotation of the material between A and C, causing both translational and rotational displacements at A. For example, if joint C rotates clockwise, joint A is forced upward, thus affecting the guy tension. Also, because of the eccentricity between A and C, the vertical guy reaction at joint A translates into both a vertical force and a moment at C, the moment arm being equal to the pole radius h. An alternate approximate way to account for the guy connection eccentricity is to assume that the guy is connected to the pole centerline at the higher point B (Fig. 2-3.a). This is discussed in the ASCE Manual for the design of guyed transmission structures (ASCE, 1997).

If joint A in Fig. 2-3.b is modeled as a joint tied to C (i.e. with the same degrees of freedom as those of C), then both A and C move and rotate by identical amounts. Even though A and C are not physically located at the same point, the modeling causes them to translate and rotate by the same amount. This is of course not exact, but it may be an acceptable approximation if the eccentricity h is small. Similarly, modeling the connection of Fig. 2-2 with a master and a tied joint ignores the small eccentricity between the two.

A master joint cannot be slave or tied to another joint.

2.3 Elements or members

Five types of elements are available in **SAPS**. These elements have been found to represent the great majority of elements needed to model realistic transmission and communication systems.

2.3.1 Truss element

The 3-dimensional truss element (or bar element) is a straight member which can only be stressed in compression or in tension. It is assumed pin-connected at both ends and has therefore a maximum of three degrees of freedom at each end. The element tension force, F, is proportional to its elongation, e. The elongation is the difference between the current length of the element and its unstressed length. The force acts in the direction of the element as shown in Fig. 2-4.

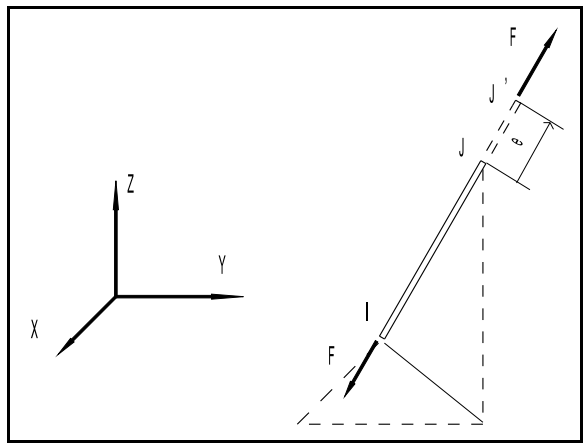


Fig. 2-4 Truss element

The unstressed length of each truss element at the reference temperature of zero degree is calculated from the input prestress value assumed to exist in the initial unloaded configuration at the input temperature TEMPIN. The input configuration is that defined by the input positions of all the joints.

2.3.2 Beam element

The 3-dimensional beam element is a straight member which can be stressed axially, in torsion, bending and shear. Figs. 2-5 and 2-6 show a typical beam element and the sign conventions used. More information can be found in the references (Au, 1993; Bathe, 1982; Hoit, 1995; Tezcan, 1969; Wang, 1983). The stress resultants and deformations at the ends of the beam element are referred to its local axes (x, y, z). The local y-axis coincides with the centroidal axis of the beam, while the local x-axis corresponds to one principal axis of inertia and the local z-axis corresponds to the other principal axis of inertia. While the determination of shears, moments and axial forces does not require that the beam be doubly symmetric, the stress check made by **SAPS** assumes a doubly symmetric cross section,

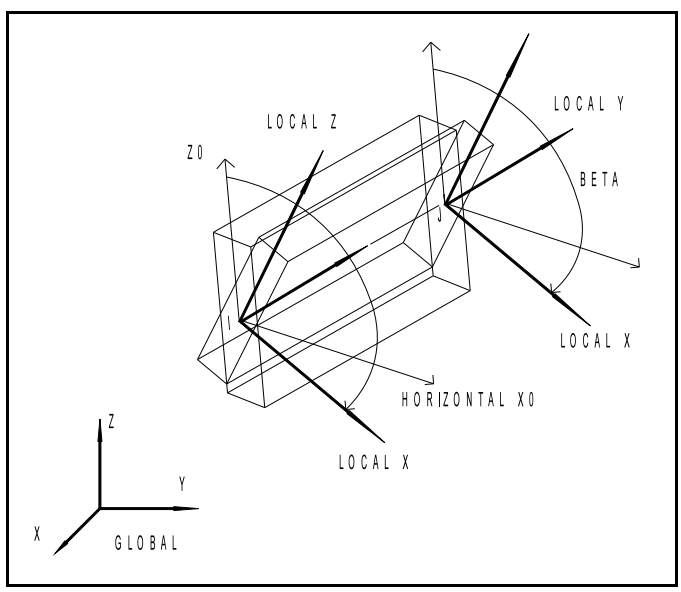


Fig. 2-5 Beam element

meaning that the principal axes are assumed located half-way between opposite sides of the beam.

The locations of the beam end joints I and J are not sufficient to completely describe the position of the beam. They are only capable of defining the position of the local y-axis. An additional quantity, the angle BETA defined in Figs. 2-5 and 2-6, is necessary to complete the positioning of the beam. For any general position of the principal axes, it is always possible to rotate the beam about its local y-axis such that the local x-axis becomes horizontal, i.e. parallel to the global XY plane, while the local z-axis makes an angle smaller than 90 degrees with the global Z-axis. This "special" position of the beam axes (shown as x0 and z0 in Fig. 2-5) is another set of axes connected to the beam element. The angle BETA, measured between the positive direction of the "special" beam axis z0 and the positive direction of the local x-axis, is used to define the position of the beam around its centroidal axis. BETA is positive as shown in Fig. 2-5.

For a perfectly vertical beam member, the orientation of the "special" beam axes is not defined by the procedure described in the previous paragraph. In such cases, it is assumed that the angle BETA is measured in the horizontal plane as the angle between the negative global Y-axis and the local x-axis in such a way that the angle between the global X-axis and the local x-axis is, in absolute value, less than 90 degrees. Fig. 2-6 illustrates the condition.

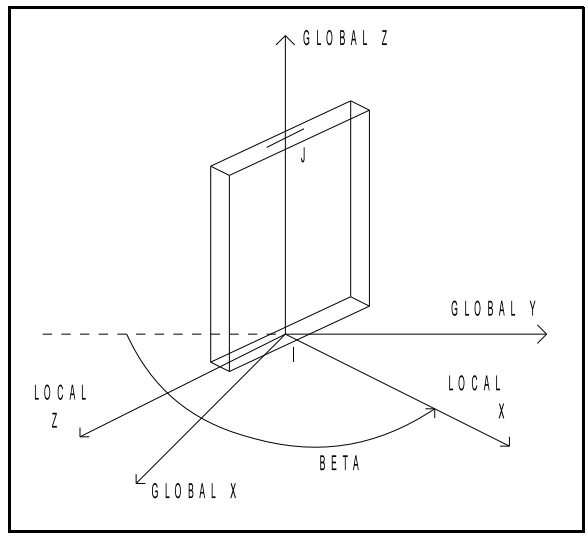


Fig. 2-6 Vertical beam

The beam element in **SAPS** allows for shear deformation. The shear deformation is equal to $V*L/G*A_V$, where V is the shear force, L is the element length, G is the shear modulus of elasticity and A_V is the shear area.

With **SAPS** nomenclature, bending about the x-axis causes deformations in Plane 1 (plane formed by the local y- and z- axes). Bending about the z-axis causes deformations in Plane 2 (plane formed by the local y- and x- axes). The printed output shows the shears and end moments in Planes 1 and 2. Fig. 2-7 shows the directions of positive axial force, shear force, torsional moment and bending moments at the origin and end of the beam.

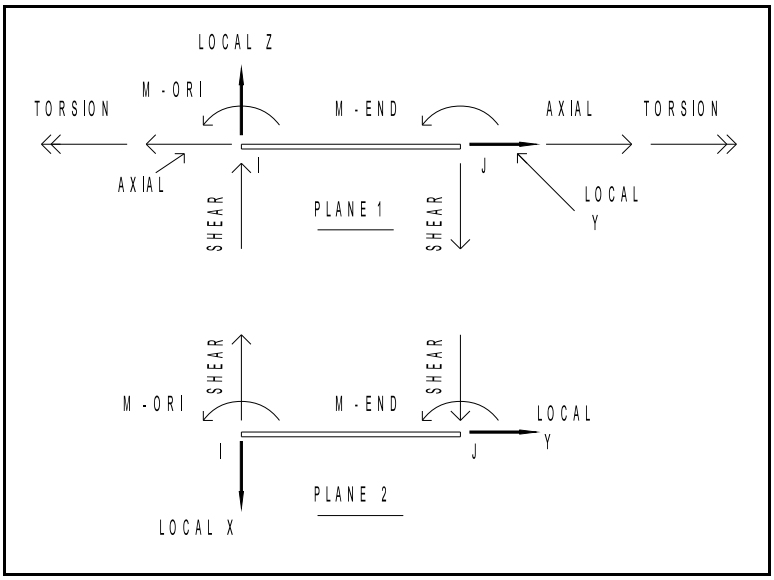


Fig. 2-7 Positive forces and moments

It is emphasized here that the beam forces shown in Fig. 2-7 are defined relative to the local orientation of the element. For example, Fig. 2-8 shows the equilibrium configuration of a simple structure that consists of only one cantilevered beam element. The only load is a vertical load of 65 kips at the end of the element. The nonlinear analysis shows a tip displacement of .2 ft, an axial force of 0.108 kips (where it appears that it should be 0) and a shear of 65 kips. The 0.108 axial load listed by **SAPS** for the element AB is the correct component of the transverse load in the direction of AB', and it is not the component in the direction of AB.

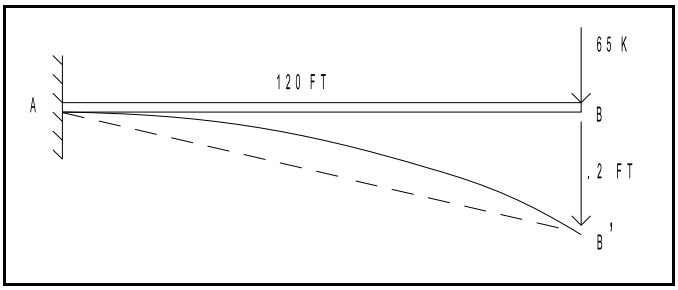


Fig. 2-8 Single cantilever beam element

Dead weight, wind and ice loads are automatically calculated for any line element, i.e. truss, beam, cable or fuse as discussed in Section 2.4. These element loads are distributed along their length. For the cable element, they are accurately modelled by distributed loads. However, for the truss, beam and fuse elements, any distributed load " w " is replaced by equivalent concentrated forces, wL/2, at each element end as shown in Fig. 2-9. Therefore the shear is always constant along the length of a beam element. Thus shear diagrams are constant and moment diagrams are straight lines between the beam ends.

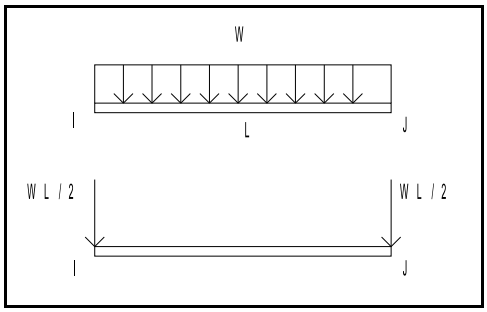


Fig. 2-9 Equivalent loads

If there is a fixed support at one end of a beam element, half of the total element load tributary to that end goes directly into the support and is not carried by the structure. For example, the shear calculated by **SAPS** in element AB of the cantilevered beam in Fig. 2-10 is only 30 kips. 10 kips go directly into the support and do not contribute to the bending and shear of the system. If AB had been shorter, the shear in it would have been more accurate.

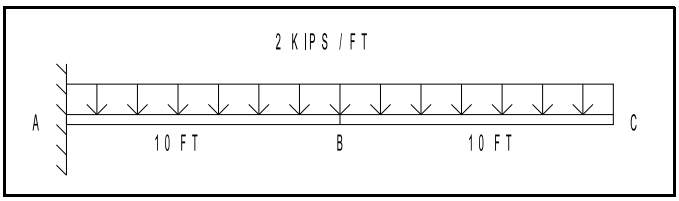


Fig. 2-10 Beam with two elements

2.3.3 Cable element

SAPS uses a proven 3-dimensional cable element capable of representing a cable from totally slack to extremely taut configurations. The element was first described in 1979 (Peyrot, 1979) but has been improved since. Unlike most cable elements used in commercial finite element programs, our cable element is an exact element, i.e. its geometry is that of an actual cable subjected to the uniform load resulting from dead weight, ice accretion and wind forces. The calculation of the resultant uniform load is presented in Section 2.4. An efficient subroutine determines the six end forces and the corresponding 6x6 stiffness matrix when the cable is subjected to various combinations of wind, ice and temperature (Fig. 2-11).

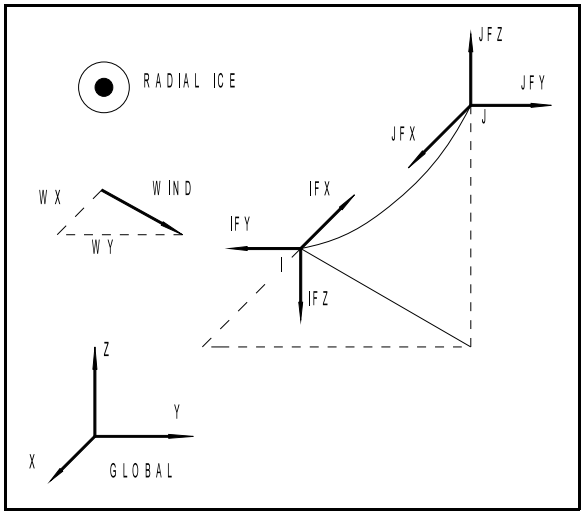


Fig. 2-11 Cable element

2.3.4 Fuse element

The 3-dimensional fuse element is a combination of an elasto-plastic truss element and a load releasing element. It is useful to model components that may yield or break. For any given load case, the fuse element may be activated or de-activated. When de-activated, the fuse element behaves just like a 3-dimensional truss element. When activated, the relationship between the tension force, F, and the elongation, e, is as shown in Fig. 2-12. When the tension elongation reaches the yield value " ety ", the force

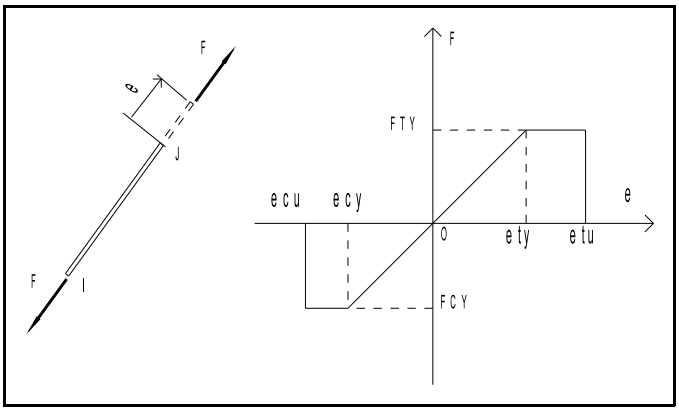


Fig. 2-12 Fuse element

ceases to increase and remains at the constant value " FTY ". When the elongation exceeds the ultimate value " etu ", the fuse releases and the force becomes zero.

The element can also yield and fail in compression as depicted in Fig. 2-12, where the shortening at yield and that at failure, " ecy " and " ecu ", are negative quantities.

It is also possible to make a beam element act as a fuse in its axial direction, but keep its bending strength. While such behavior does not exist in actual structures, the option can be used in our **TOWER** program (which uses **SAPS** as its analysis engine) to take care of some special unstable mechanism problems.

Important Note: While the fuse element may be specified with strength and deformation limits in both tension and compression, it is not recommended to use limits in tension and compression simultaneously. This may result in numerical instability, as the status of the element may forever change between tension and compression. One practical way to avoid the problem is to use a very large fictitious yield force " FCY " and a very large ratio " ecu " / " ecy " if the fuse is to be used in tension, and conversely to use very large limiting values in tension if the fuse is to be used in compression.

2.3.5 Linear substructure

A linear substructure (or super-element) is any portion of the system which has known stiffness properties at its attachment points to the overall system. Substructures are tied to the system at their external joints only. All internal joints have been condensed out (Au, 1993; Bathe, 1982; Cook, 1981).

A substructure is entirely defined by its external joints and the stiffnesses at these joints. For example, we may only be interested in the locations and longitudinal stiffness properties of the five ground wire and insulator attachment points A, B, C, D and E on the H-frame of Fig. 2-13. The relationships between the longitudinal loads in the five directions 1 to 5 and the corresponding longitudinal displacements in the same five directions are described by a 5x5 substructure stiffness matrix. The properties of the H-frame are said to be condensed out to the five attachment points in the longitudinal direction.

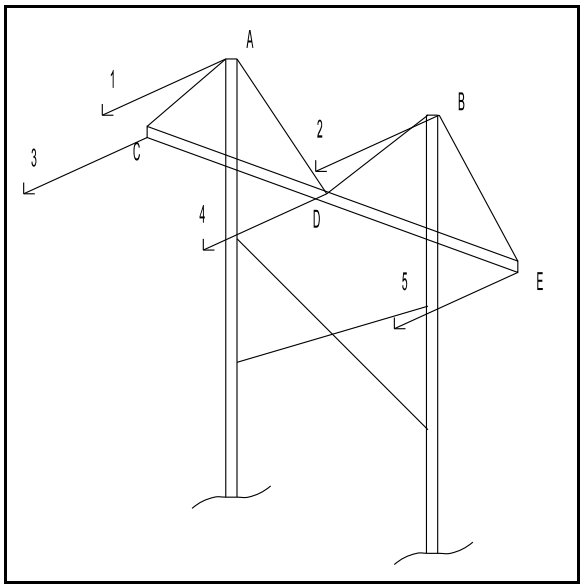


Fig. 2-13 H-frame as a substructure

Each substructure joint, like any joint at the ends of truss, beam, cable or fuse elements, may have from 1 to 6 DOF's. The substructure DOF's are numbered

from 1 to NDFSUB, where NDFSUB is the total number of degrees of freedom of that substructure. This numbering system is the local DOF numbering system. The substructure stiffness matrix is of size NDFSUBxNDFSUB. For example, in the frame of Fig. 2-13, because we are only interested in the longitudinal displacements of the attachment points, we only need to consider one DOF per attachment point. Therefore, NDFSUB is equal to 5 and the stiffness matrix is 5x5. Had we been interested in all three components of displacements at all attachment points, NDFSUB would have to be equal to 15. Example 5 in Section 5 shows how **SAPS** can automatically generate the stiffness matrix of a structure condensed to its cable attachment points.

When a substructure is tied to the overall system, for example when the frame of Fig. 2-13 is tied to other frames through conductor and ground wires, the local substructure DOF's coincide with global system DOF's. The mapping of the local DOF's onto the global DOF's has to be described.

Substructures can only be used as condensed models of linear structures. They can be used as parts of larger nonlinear systems, but the structure underlying each substructure has to be linear. Using substructures is a convenient way to reduce the size of large problems as demonstrated by Example 4 in Section 5. The same concept is used in our **PLS-CADD** program when segments of lines are modeled internally by **SAPS** at Level 3 (see Section N.3 of the **PLS-CADD** manual for details).

2.4 Loads

This section discusses how loads are handled by **SAPS**. Some loads are calculated automatically, such as temperature, dead, ice and wind loads on truss, beam, cable and fuse elements, while others have to be described as input data.

2.4.1 Dead, temperature and ice loads

Loads produced by dead weight, temperature change and ice accretion are automatically calculated for the truss, beam, cable and fuse elements.

2.4.2 Wind (or current) loads

Truss, beam, cable and fuse elements can also be subjected to wind loads (or current loads in offshore applications). The global wind is defined by its profile (increase of velocity with height) and its direction (angle ALPHA measured from the global X-axis) as shown in Fig. 2-14.

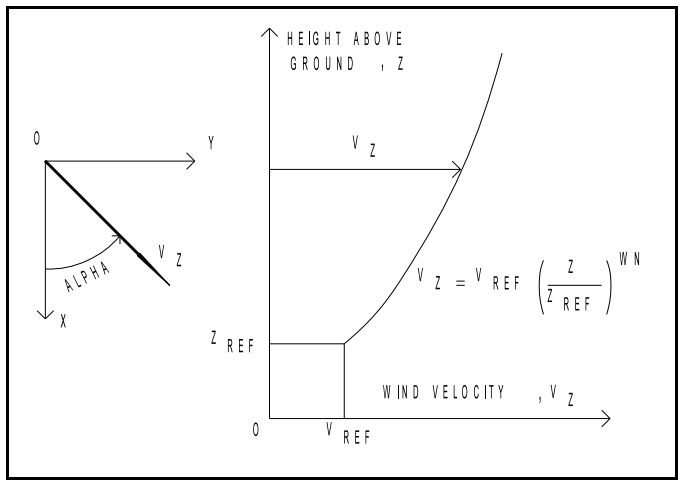


Fig. 2-14 Global wind profile

You select the type of profile in the dialog box of Fig. 4.14-1. The profile can be “exponential” as shown by the equation in Fig. 2-14, or “logarithmic”, or “quadratic”, or it can be defined at discrete points of different heights above the ground. When the profile is defined at discrete points, it can be constant between these points (if you select the “step” option) or it can vary linearly. This is described in Section. 4.4.4.

If you select the “exponential” profile, the wind velocity V_Z at height z above the ground is related to the reference wind velocity V_{REF} at the reference height Z_{REF} (usually 30 ft or 10 m) by the equation shown in Fig. 2-14. The power law for the global wind velocity profile in Fig. 2-14 is used in many codes and design guides (ANSI/ TIA, 2005; ASCE, 1991; ASCE, 2002; TIA/ EIA, 1996) where the power coefficient WN is often taken as a number between 1/12 and 1/5.

It is assumed that the wind pressure acting on each element is constant along the length of the element and equal to that caused by the wind at mid height of the element. The magnitude of the force produced by the wind on a unit length of element is calculated by the formula:

$$w_w = .5\, \rho\, C_d\, d_e\, (V_{zn})^2 \qquad \text{Eq. 2-1}$$

where:

ρ = mass density of moving fluid (air or water)
C_d = drag coefficient for element
d_e = outside diameter or exposed area per unit length (including ice coating if any)
V_{zn} = component of global wind velocity at midheight of element in direction normal to the chord of the element

The force vector w_w is in the plane formed by the chord of the element and the global wind velocity vector, and it is directed in the direction normal to the chord. Eq. 2-1 gives results similar to the EIA method for computing wind loads on guys (Peyrot, 1987; TIA/ EIA, 1996) if $C_d = 1.2 \times G_H$, where G_H is the tower gust response factor.

2.4.2.1 Special override for cable elements

In order to simulate some code specifications that requires that you apply a gust response factor and a wind perpendicular to each cable element (local wind) regardless of the direction of the global wind, the following alternate to Eq. 2-1 is available as an option for cable elements only:

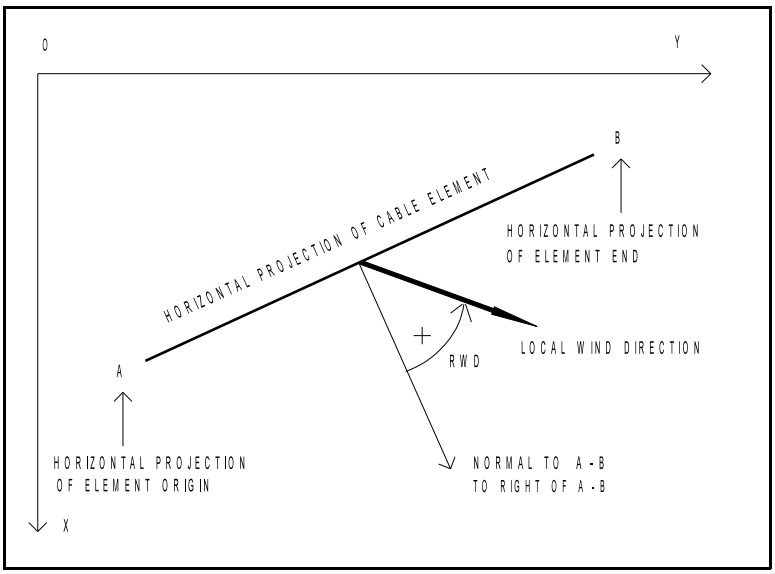


Fig. 2-14.a Local wind direction

$$w_w = GRF \times \{ .5\, \rho\, C_d\, d_e\, (V_{zn})^2 \} \qquad \text{Eq. 2-1.a}$$

where:

GRF = cable gust response factor
V_{zn} = component of local wind velocity at midheight of cable element in direction normal to the chord of the element

The force vector w_w is in the plane formed by the chord of the cable element and an horizontal local wind velocity vector having the magnitude of the global wind at mid-height of the element and a local direction relative to the horizontal projection of the element defined by the angle RWD shown in Fig. 1-14.a. RWD is positive as shown. In most code applications, RWD is equal to zero or 180 degrees.

2.4.3 Total distributed load on element

For truss, beam, cable and fuse elements, the resultant load vector per unit length, w_t , is the vector sum of the gravity load w_v (dead plus ice) and w_w (Eq. 2-1), i.e. $w_t = w_v + w_w$. The algorithm for cable elements assumes that w_t is applied uniformly along the entire length of the cable. However, for truss, beam and fuse elements, the resultant load is replaced by a pair of concentrated forces at the ends of the element, each equal to w_t times one half the length of the element.

2.4.4 Concentrated forces or moments

Concentrated forces and moments can be applied at any joint and in any direction. They are simply specified by their components in the global coordinate system.

2.4.5 Specified displacements or rotations at supports

Displacements or rotations of fixed joints (supports) can be specified in any direction, i.e. a direction that does not have a degree of freedom associated with it.

2.5 Solution strategies for nonlinear problems

Solving highly non-linear problems is much more difficult than solving linear or slightly non-linear problems. Many methods have been described in the literature, each having its advantages and disadvantages. The methods that can be used with **SAPS** are modified Newton-Raphson techniques.

2.5.1 Basic Newton-Raphson method

The basic Newton-Raphson method works well for slightly nonlinear systems. The method can be explained with the single-degree-of-freedom model shown on the left of Fig. 2-15. IJ represents the unstretched position of a nonlinear bar fixed at I and supported on roller at J. The goal is to find the equilibrium position of the system (i.e. the displaced location of J at J') when a force P is applied at J. Curve OFDBE on the right of Fig. 2-15 shows the slightly non-linear relationship between the tension force in the bar, F, and its elongation e = X - XI.

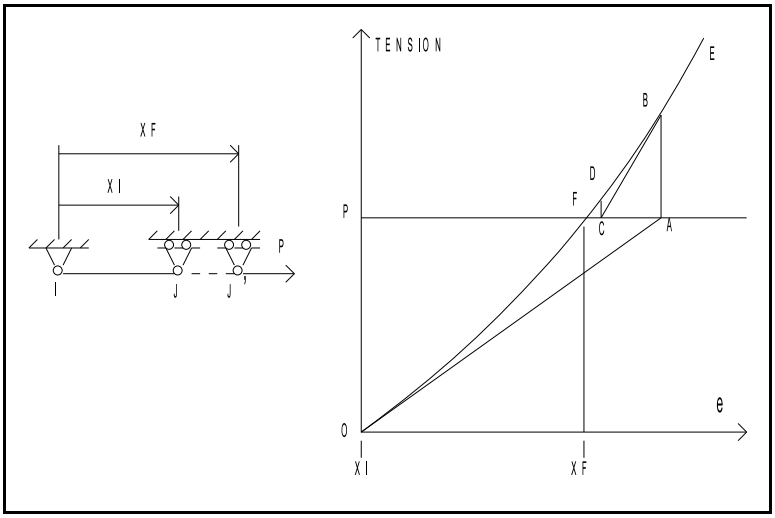


Fig. 2-15 Basic Newton-Raphson method

At any step of the iterative procedure used to find the location of point J', there exists an imbalance of force, Q, at the end of the bar. Q is equal to P - F. One tries to get rid of this imbalance by making a displacement correction $\Delta X = Q / K_t$, where K_t is the tangent stiffness, or the slope of the F - e curve at the point where the correction is being made. For example, at the first iteration, Q is equal to OP, K_t is the slope of line OA and PA is ΔX. At the second iteration, Q is equal to BA, K_t is the slope of line CB and AC is ΔX. As can be seen from Fig. 2-15, the procedure quickly converges toward the solution. That procedure is the well known Newton-Raphson technique.

2.5.2 Modified Newton-Raphson method with stiffness addition

Flexible systems, especially those that include slack cables, may be highly nonlinear. In such cases, the basic Newton-Raphson method may not converge toward a stable solution. This can easily be visualized in Fig. 2-16 which represents an initially slack cable. The problem of Fig. 2-16 is identical to that of Fig. 2-15, except that a cable element replaces the slightly non-linear bar. The relationship between the cable chord length and its tension is highly non-linear as depicted by curve OADGKF in Fig. 2-16. This is so because the cable goes from slack to taut as its end J moves to the right. It can be seen that the Newton-Raphson method would not converge for this problem as the displacement correction would be much too large at the first and possibly subsequent iterations.

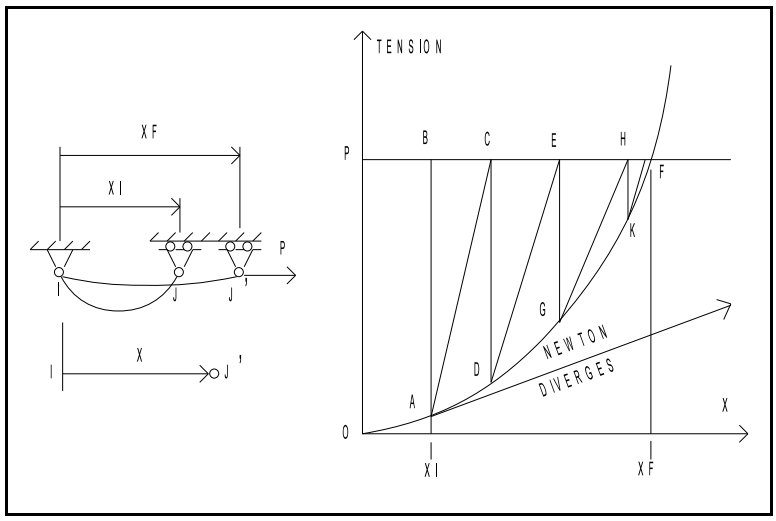


Fig. 2-16 Modified Newton-Raphson method

Divergence occurs because the cable tangent stiffness (slope of tangent to tension vs. elongation curve) is too small. By artificially increasing that stiffness, it is possible to bring the iteration process under control. The stiffness is increased at the time the displacement correction is calculated. Therefore, the displacement correction is $\Delta X = Q / (K_t + K_A)$, where K_A is the additional stiffness needed to insure convergence.

As can be seen from Fig. 2-16, an efficient solution algorithm is one where K_A is initially large but becomes small as the solution nears the equilibrium point. **SAPS** allows the addition of a variable stiffness correction term at each iteration. The form of the stiffness correction term is:

$$K_A = KM + KV / (i^{PW}) \qquad \text{Eq. 2-2}$$

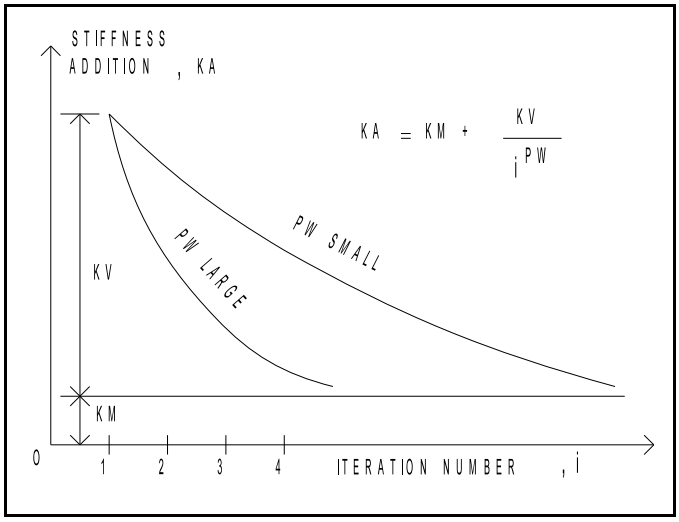


Fig 2-17 Stiffness correction

where KM is the minimum correction term, KV is the maximum variable correction term, i is the iteration number and PW is the power at which i is raised. With the proper choice of the 3 stiffness correction parameters KM, KV and PW, almost any stiffness correction scheme can be achieved. Fig. 2-17 shows qualitatively the effect of varying the stiffness parameters. With KM and

KV both equal to zero, the method reverts to the basic Newton-Raphson method.

If, instead of the single degree of freedom structure of Fig. 2-16, one needs to analyze a highly non-linear problem with many degrees of freedom, the scheme described above can be generalized (Peyrot, 1979). The search for static equilibrium involves displacement corrections of the type:

$$\{X^{i+1}\} = \{X^{i}\} + [\,[K^{i}] + K_A\,]^{-1}\ \{Q^{i}\} \qquad \text{Eq. 2-3}$$

where

i	=	iteration number
$\{X^i\}$	=	vector of joint positions at step i
$\{Q^i\}$	=	vector of joint imbalances at step i
$[K^i]$	=	system tangent stiffness matrix at step i
K_A	=	diagonal matrix with all diagonal elements equal to the expression of Eq. 2-2

2.5.3 Newton-Raphson in combination with load increment

Whether one uses the basic Newton-Raphson method or the modified method with stiffness addition (Eq. 2-3) , convergence to the solution is always easier if the equilibrium position is not too distant from the starting position. This fact can be used to advantage by performing successive analyses under increasing loads. Instead of applying the full load to the system in a single load case, the load is progressively increased at each analysis. The equilibrium configuration at the end of an analysis becomes the starting configuration at the start of the next analysis. Fig. 2-18 illustrates the incremental load concept. P_i is the load applied during the i-th analysis (i.e. the i-th load case). **SAPS** as well as most non-linear finite element programs support the load increment option.

With **SAPS**, it is possible to use the stiffness addition method as well as the load increment method concurrently. For every load case, the user has the choice of starting the search process from the initial unloaded system configuration or from the equilibrium configuration of the previous load case.

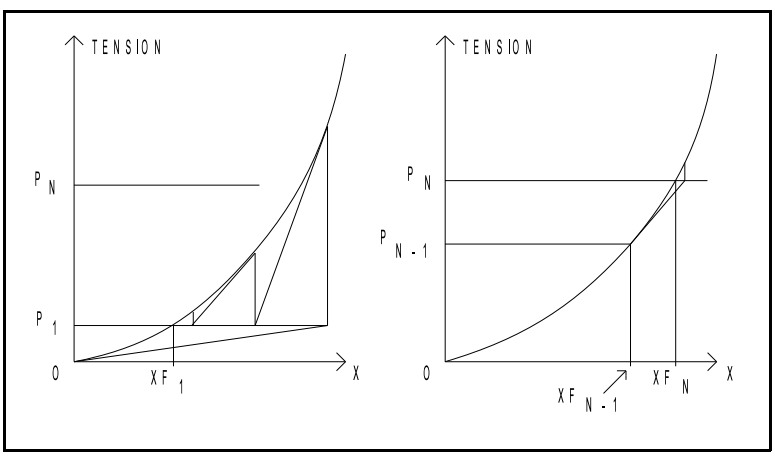


Fig. 2-18 Load increment strategy

2.5.4 Ill-conditioned stiffness matrix

Systems with cable components (or cables modeled by series of truss elements) often exhibit an ill-conditioned stiffness matrix, i.e. a matrix that cannot be inverted numerically due to division by zero or loss of accuracy. Ill-conditioning may develop at a joint between two cable or truss elements in series. At such a joint, there is considerably more stiffness in the general direction of the elements (in the plane of the elements) than in the perpendicular direction. The geometric stiffness, which is proportional to the tension in the elements, is the only stiffness that exists in the direction perpendicular to the elements. There is usually no problem with taut cables, which have substantial out-of-plane geometric stiffness. However, at some stage of the calculations a cable may become slack and loose its geometric (lateral) stiffness. In such a case, ill-conditioning occurs.

The stiffness correction term in **SAPS** which was described in Section 2.5.2 as a mean of controlling convergence can also be used to avoid an ill-conditioned global stiffness matrix. The additional stiffness provided at the joints which have insufficient stiffness may remove the ill-conditioning problem.

2.6 Strength checks

SAPS is basically an analysis program. For each load case, it determines the forces or moments in each element. In addition, it has some limited design check reporting capabilities if the proper strength data are given as input. However, if you need to make extensive strength checks, especially some that are code-dependent, you should use our **TOWER** or **PLS-POLE** programs.

The design check for each element is given in the form of a percent of strength used, where that strength usage (or capacity utilization) is printed in the **Analysis Results** report (see Section 3.4) or is displayed as a number or color for each element in the **Deformed Geometry** window. There is no strength checks for substructures.

2.6.1 Truss, cable and fuse elements

For truss and fuse elements, strength usage in percent (SU) is given by the largest of the two ratios:

SU = Maximum tension force / *Tension capacity* Eq. 2-4

or SU = Maximum compression force / *Compression capacity* Eq. 2-5

For cable elements, strength usage is defined by Eq. 2-4.

2.6.2 Beam elements

For beam elements, strength usage is calculated according to one of the following options. The option type and the required strength data (shown in italic in Eqs. 2-6 to 2-9) are defined for each beam element in the **Geometry/ Beams** menu.

2.6.2.1 Axial check only

This check may be appropriate when angle members in a steel latticed tower are modeled as beam elements to avoid unstable planar joints or mechanisms, yet they basically behave as tension/compression truss members. In such cases, the check made is that of Eqs. 2-4 and 2-5.

2.6.2.2 Stress check #1

With this option, SU is given by the equation:

$$SU = (P / A + M_X C_X / I_X + M_Z C_Z / I_Z) / \textit{Allowable normal stress} \qquad \text{Eq. 2-6}$$

where: P = Absolute value of axial load

A = Cross section area

M_X and M_Z	=	Absolute values of moments about local X- and Z- axes
C_X and C_Z	=	Distances from neutral axis to edge of beam in directions perpendicular to local X- and Z- axes
I_X and I_Z	=	Moments of inertias about local X- and Z- axes

This check may be appropriate for a beam with rectangular or I cross section.

2.6.2.3 Stress check #2

With this option, SU is given by the equation:

$$SU = (P / A + SQRT [M_X M_X + M_Z M_Z] C_X / I_X) / \textit{Allow. normal stress} \qquad \text{Eq. 2-7}$$

where the variables are the same as those in Eq. 2-6.

This check may be appropriate for a beam with circular or near-circular cross section. In such a case I_X has to be equal to I_Z and C_X has to be equal to C_Z.

2.6.2.4 Interaction check #1

With this option, SU is given by the interaction equation:

$$SU = P / \textit{Axial capacity} + M_X / \textit{Moment capacity about X-axis} + M_Z / \textit{Moment capacity about Z-axis} \qquad \text{Eq. 2-8}$$

where:

P	=	Absolute value of axial load
M_X and M_Z	=	Absolute values of moments about local X- and Z- axes

This check may be appropriate for a beam with rectangular or I cross section.

2.6.2.5 Interaction check #2

With this option, SU is given by the interaction equation:

$$SU = P / \textit{Axial capacity} + SQRT [M_X M_X + M_Z M_Z] / \textit{Moment capacity about X-axis} \qquad \text{Eq. 2-9}$$

where the variables are the same as those in Eq. 2-8.

This check may be appropriate for a beam with circular or near circular cross section which has the same moment capacity regardless of direction.

3. LOADING, RUNNING AND VIEWING EXISTING MODEL

This section describes how to load, run and view an existing example. It is assumed that you are familiar with the basic MS-WINDOWS commands and terminology. To get started, double click on the **SAPS** icon. You will be taken to **SAPS** main menu with the **File** menu at the upper left corner.

3.1 The File menu

In this section, we will use the guyed tower of Example 1 to illustrate the capabilities of **SAPS**. To load Example 1, click on **File/ Open**. The dialog in Fig. 3.1-1 will appear. **SAPS** models are generally saved in files with the *.sap* extension. Double click on the *ex1.sap* icon to load the Example 1 model. The initial geometry of the model will appear as shown in Fig. 3.1-2. We will discuss that model further in Section 3.2. Between here and Section 3.2 you will find information regarding other items under the **File** menu. You can browse through these items but do not take any action that might affect Example 1.

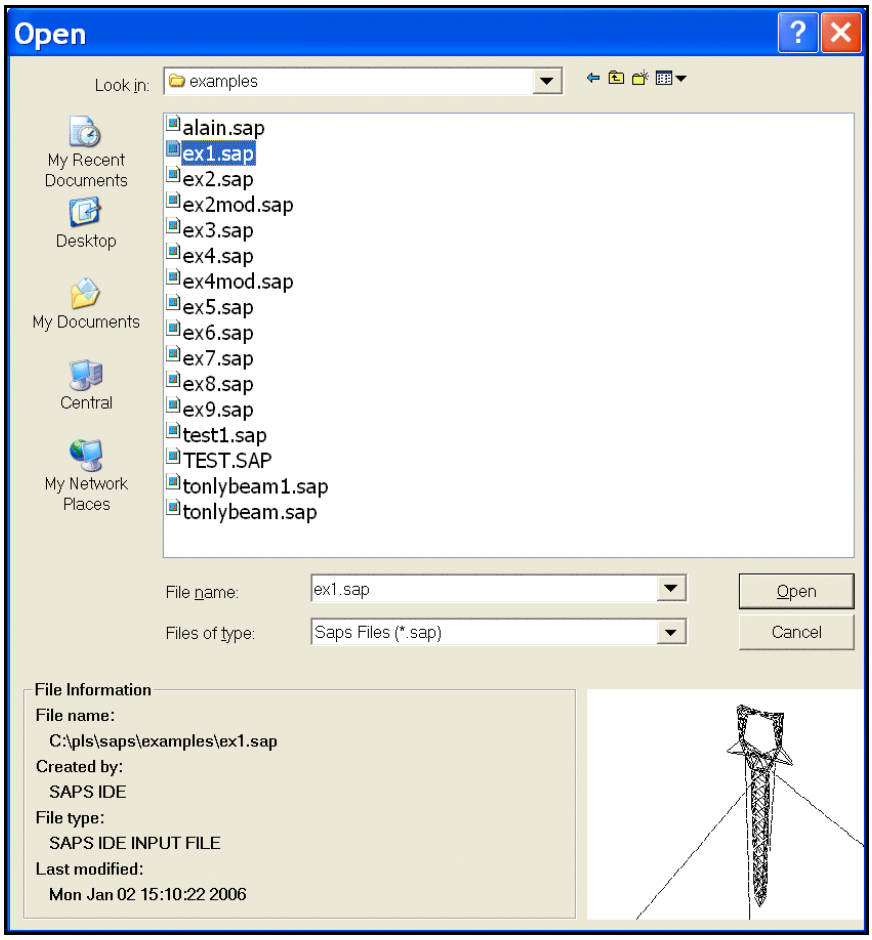


Fig. 3.1-1 Files dialog box

If you were building a new model, you would select **File/ New** instead of **File/ Open** and you would fill-in the data in the menus of Section 4. But do not attempt to build a new model at this time until you have gained experience with the examples which are provided.

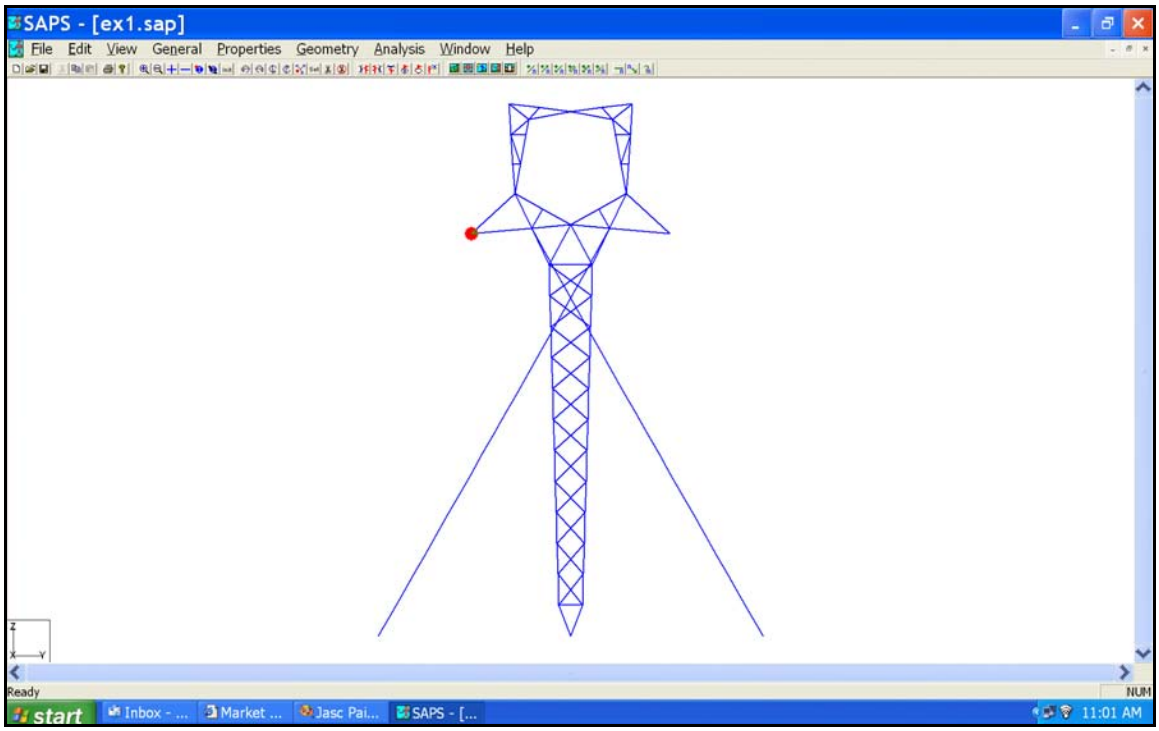


Fig. 3.1-2 Example 1 model

3.1.1 Directory preferences, selection of component files and units

To select units, component libraries (files), and other preferences, click on **File/ Preferences**. You will be taken to the **Preferences** dialog of Fig. 3.1-3 where you can select the following:

Unit system: This is where you specify which unit system to use. You can select between US customary, SI/Metric or consistent

Report font: Font to be used in all text windows and reports

Table font: Font to be used in all input tables

View font: Font to be used for all text in the **Initial** and **Deformed Geometry** windows

View background color:

Clicking on this button will allow you to select the background color of all your view windows

Then you need to specify the default directories and files for new projects. These are the directories and files which will be used after you select the **File/ New** command.

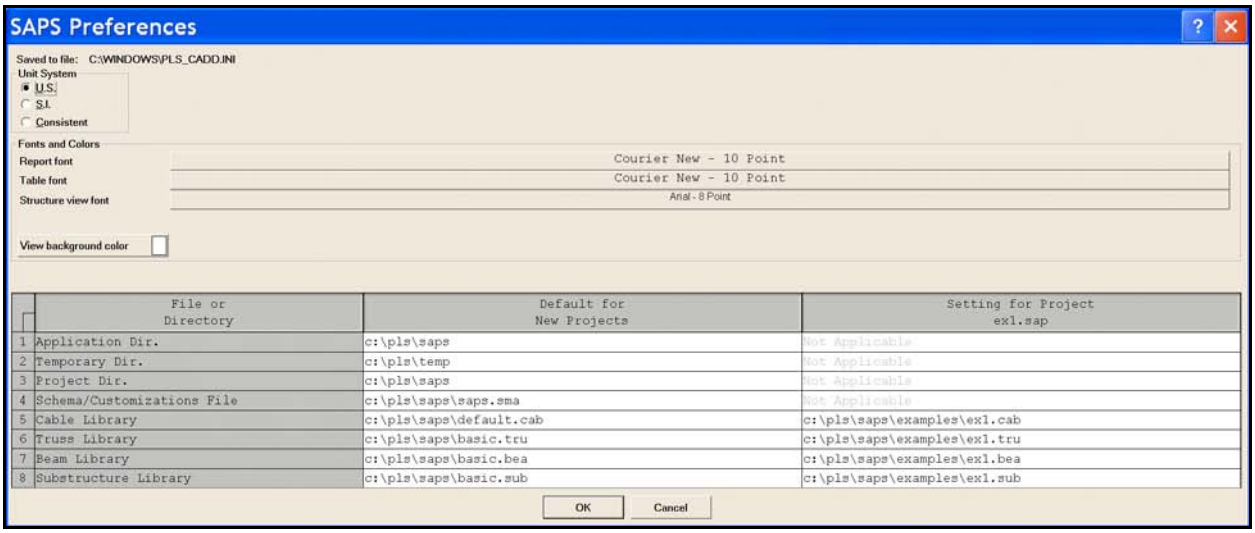


Fig. 3.1-3 Preferences dialog box

Application directory: This is where the **SAPS** executable resides. It must be specified correctly or other programs (like **PLS-CADD**) will not be able to link to **SAPS**.

Temporary directory: This is where all temporary files are written: **Important note**: the *Temporary Directory* should be specified on your local computer, even if you are working with files on a network. This will prevent time lost accessing the network and the possibility of collisions with others trying to access the same directory

Project directory: This is where the geometric model of your structure (as built with the various instructions in Section 4) is stored by default. This is also used as the root for the files listed under **Setting for Project** column

Schema and Customizations File:

This is where the information that allows you to customize menus/ dialog boxes and tables/ reports (see Appendix D) is stored. This file must have the "*.sma*" extension. If you have a "*.sma*" file for a different language, you may select it here.

Components libraries:

The *Cable*, *Truss*, *Beam* and *Substructure Libraries* are the names of the files (also referred to herein as databases or libraries) which contain the properties of the various components used to build your structure model. These components are described in Section 4.2. Component libraries must have the appropriate extensions: "*.cab*", "*.tru*", "*.bea*" or "*.sub*", respectively.

When you OK the **Preferences** dialog, the *Default for New Projects* settings are automatically saved in the file *PLS_CADD.INI* in your *WINDOWS* directory and remain in effect until changed.

Finally, you need to specify the files (databases or libraries) where the components used for a particular project (actual structural model) reside. These files are usually the same as those selected for new projects, but they need not be. The file names for *Cable*, *Truss*, *Beam* and *Substructure Libraries* to be used in connection with your current project are specified in the *Setting for Project* column of the **Preferences** table. Note that this column is only available when you have a project loaded. When you OK the **Preferences** dialog, the *Setting for Project* information, which really includes pointers to appropriate component libraries, is saved together with the corresponding structure geometry information.

3.1.2 Saving or backing-up model

A **SAPS** model is made up of its geometry and loads. It is stored in a **.sap* file. The model also includes references to particular element types whose properties are included in separate libraries: **.tru* files for truss/ fuse elements, **.bea* files for beam elements, **.cab* files for cable elements and **.sub* files for substructures.

When selecting **File/ Save**, you are saving the model currently residing in memory to a designated file, say *Model.sap*. **File/ Save** has no effect on the element *Properties files* which the model refers to.

Occasionally, you might want to save in a single file, say *Model.bak*, the model as well as the element property files that the model refers to. Therefore, *Model.bak* is a complete record of the information available at the time a model is backed-up. That information can be restored on the same or another computer with **File/ Restore backup**. *Model.bak* includes not only files but the complete directory structure. Upon restoration, the entire file and directory structure will be recreated. Because the single file *Model.bak* contains all the information to recreate a model, including the element data bases, it greatly simplifies the transfer of model data from one computer to another. However, you should avoid using **File/ Backup** too often, as you might end up with many unnecessary copies of your element data bases.

When you use **File/ Restore backup**, you are given the opportunity to change the name of the directory in which the various files are kept.

3.2 Model description

When a model is loaded, it is displayed in the **Initial geometry** window, as shown in Fig. 3.1-2. The initial line of sight is parallel to the global X-axis, i.e. the model is projected onto the Y-Z plane. The **Initial geometry** window is labeled with the name of the model. Cables in the **Initial geometry** window are displayed as straight lines.

3.2.1 Viewing of model in **Initial Geometry** window

The view in the current geometry window can be modified through the use of the following commands. The commands are available under the **View** menu and/or by clicking on the appropriate toobar buttons. Using the buttons is recommended.

Zooming or Inverse Zooming

In	Click the + button on the toobar or press the + key on the keyboard
Out	Click the - button on the toolbar or press the - key on the keyboard
Window	Select **View/ Zoom Rectangle** and drag a window over the part of the model you wish to zoom on. This is done by bringing the mouse cursor to the desired location of the upper left corner of the window and dragging the cursor to the lower right corner. A magnifying glass will remind you that you are in zoom mode. To get out of zoom mode, or any other viewing mode, right-click the mouse. The **Zoom In** toolbar button is a short cut to **View/ Zoom Rectangle**.

Panning

Press the **Left**, **Right**, **Up** or **Down** arrow keys on keyboard or click the arrows of the window scroll bars.

Longitude and latitude rotations

Changing the latitude or longitude of your line of sight is done by clicking on the **Lat+**, **Lat-**, **Long+** and **Long-** buttons on the toolbar, pressing the **Pg Up**, **Pg Dn**, **End** and **Home** keys, or entering desired values in the **3-d Control** dialog box described below. The amount of change effected by each click is defined in the **Rotation Increment** field of the **3-d Control** dialog box. The origin of the line of sight, defined in Fig. 3.2-1, passes through the origin of the global coordinate system XYZ unless changed with the **View/ 3d/ Rotation origin** command.

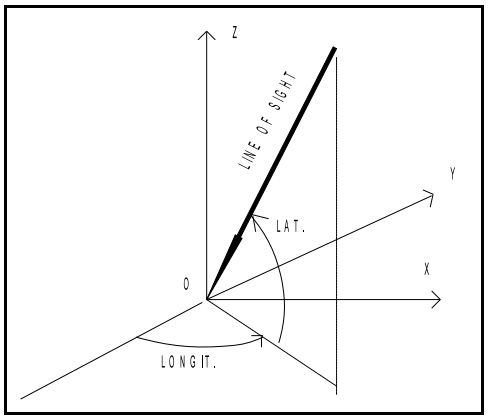


Fig. 3.2-1 Line of sight

Views with preset latitudes and longitudes of 0 or 90 degrees can be obtained directly by clicking on one of the **View from ...** buttons in the toolbar.

Eliminating portion of section from view

You can eliminate from view any portion of the model inside a "cut window". To define a cut window, click on the **Cut** button and drag the cut window over the region which you want to eliminate. Click on **Uncut** to bring back all members into view.

Restoring original view

To restore the original view, select **View/ Initial** from the main menu or click on **Init** button.

Restoring previous view

To redraw the previously drawn view, select **View/ Previous**.

Line thickness and background color

View/ Display Options/ Line Width from the main menu lets you change the thickness of all lines by specifying the number of pixels used in their representation. **View/ Display Options/ Background Color** lets you select the background color for all graphics windows.

Printing, saving or exporting view in graphics window

Use **File/Print** from the main menu to print the view in the current window. Use **File/ Export DXF** to export the view in DXF format. You can also save the view for future use as a *.plt* file by clicking on **Save as** in the menu that appears when you right click anywhere in the window (do not use the **File/ Save** from the main menu for this purpose as it will save the model rather than the graphics view). A *.plt* file can be read by any one of our programs using **Window/ New View**.

Measuring distances between points

If the latitude and longitude of your line of sight are a multiple of 90 degrees, you can measure the distance between any two points on the screen with **View/ Distance Between Points**. You click on the first point, then drag (rubber band) the mouse to the second point. The distance and its projections are displayed in the bottom status bar.

Measuring distances between joints

This is different from measuring distances between arbitrary points on the screen, as described above. With **View/ Distance Between Joints**, you can measure the distance in 3-d between any two

designated joints in your model. You click on the first joint (the mouse cursor snaps to the closest joint on the screen) to establish the origin of the measurement. This first joint label and coordinates are displayed in the status bar. Then you move the cursor close to the second joint. Its label and the distance between the two joints are then displayed in the status bar. Right click to get out of the measuring function.

Other display commands available from "3-controls" dialog box

The remainder of the display options are available in the **3-d Controls** dialog box which is opened by clicking on the **Set** button. The **3-d Controls** dialog box is shown in Fig. 3.2-2.

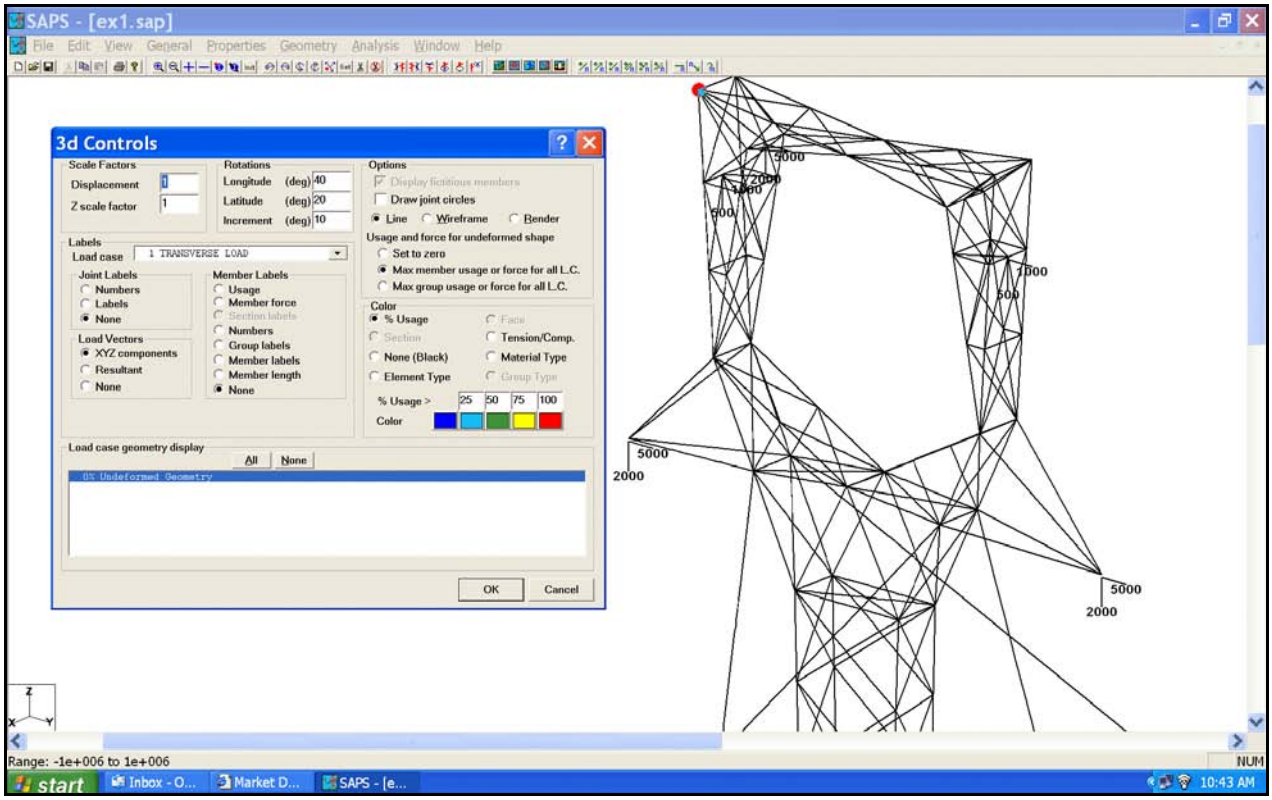


Fig. 3.2-2 3-d Control dialog box

You can selectively show or hide the label or number of all joints and members. Labels are alphanumeric descriptions assigned to the items in the input tables. Numbers are internal values assigned by the program and are generally not of interest.

When in the **Initial Geometry** window, you can display the concentrated loads for any load case by checking the *Load vectors* item and picking the desired load case in the *Labels/ Load case* box. For example, in Fig. 3.2-2, the conductor loads for the *Transverse Load* case are displayed.

There are additional controls in the **3-d Controls** dialog box that pertain to views of deformed geometries (analysis results). This is discussed in Section 3.5. The " *Wire frame* " and " *Render* "

options which require dimensional information for the members (for example steel angles in **TOWER** and tubular poles in **PLS-POLE**) are not available in **SAPS**.

3.2.2 Model creation and editing

Creating and editing a model is done through the **General**, **Material** and **Geometry** menus. The process is described in Section 4.

3.3 Running the analysis

Once a model is created, it can be checked for commonly occurring mistakes using the **Analysis/ Check** command. Once checked, the analysis is performed with the **Analysis/ Run** command. Depending on the size of the model and the type of analysis (linear or nonlinear), the analysis may take from a fraction of a second to several minutes to complete. The status bar and the hourglass on the screen tell you that the analysis is being performed.

Once the analysis is completed, one text report (labeled **Analysis Results**) and a **Deformed Geometry** window are opened. You can see all windows simultaneously with **Windows/ Tile Horizontal** or **Windows/ Tile Vertical.**

3.4 Analysis results report

The **Analysis Results** report (see Fig. 3.4-1) echo-checks all input data and shows detailed results of the analysis: joint displacements, member forces and moment, and some design checking when appropriate.

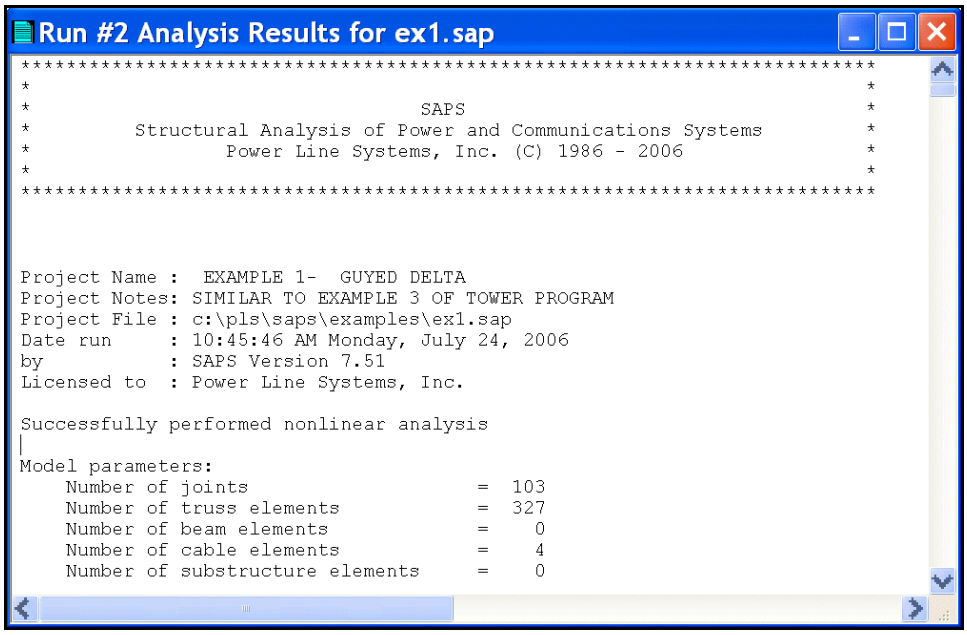


Fig. 3.4-1 Analysis results report

3.4.1 Viewing, printing and exporting

There are several useful commands that let you manipulate our text reports. To access these commands, right click the mouse while its cursor is in the text window. A small context menu will appear with the following commands:

Save or **Save as** let you save the report in the *.txt* format. That file can later be edited with a word processor which can read *.txt* formatted files and printed.

Append to lets you append the report to the end of an existing file.

Font lets you change the fonts for viewing the report window.

Autosize Font may be used to let the program select the largest size font for which none of the report lines wrap when printed. With large reports, this may take some time to execute.

If, prior to right-clicking the mouse, you select a portion of text by dragging the mouse or select the entire report with **Edit/ Select All**, then, in addition to the **Save**, **Append to** and **Font** commands described above, you can access the following commands which will operate only on the selected text:

Copy lets you copy the report to the WINDOWS clipboard. The report can then be retrieved by any word processing program (Word, WordPerfect, etc.) allowing a paste from the WINDOWS clipboard.

If, instead of sending the report to a word processing program for printing, you wish to print the report directly, you can do it with **File/ Print**, while the report window is selected.

3.4.2 Customizing

There is a limited capability in **SAPS** to customize the **Analysis Results** report. This is dicussed in Appendix G of the **TOWER** manual.

3.5 Deformed geometry window

The **Deformed Geometry** window provides you with a graphical summary of the analysis results. The picture that you see depends on the selections made in the same **3-d Control** dialog box which was first described in Section 3.2.1 for the **Initial Geometry** window. That **3-d Control** dialog box is opened by clicking on the **Set** button.

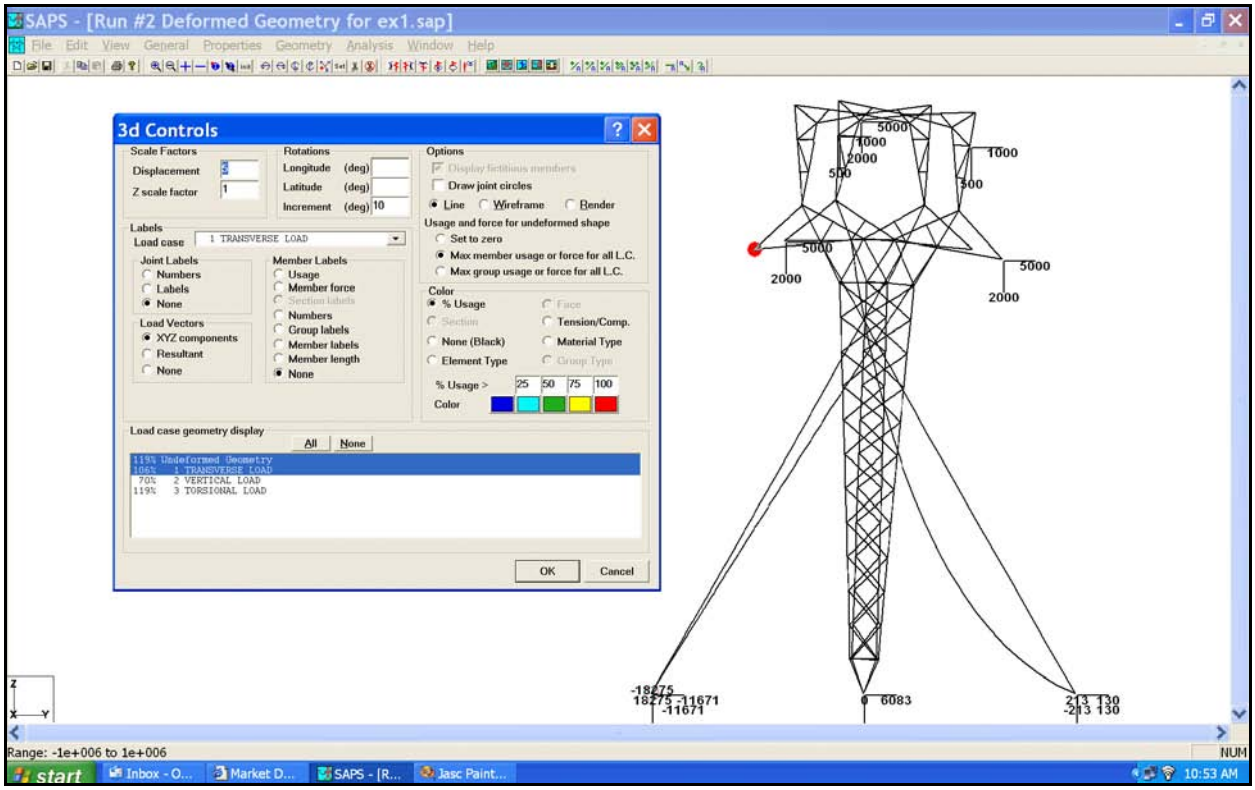


Fig. 3.5-1 3-d Control box in Deformed Geometry window

When it is first opened, the **Deformed Geometry** window displays, at scale, color coded views of the undeformed structure and its deformed configurations under all load cases simultaneously.

The following options are available the **3-d Controls** box:

Viewing one or more deformed shapes simultaneously

This is done by selecting the *Undeformed Geometry* or any number of *Deformed Geometries* (identified by load case titles) in the **Load case geometry display** portion of the **3-d Control** box.

Amplifying the deflections of deformed shapes

You can amplify the displacements of the joints of all displayed deformed shapes by entering a **Displacement Scale Factor** larger than one. For example, in Fig. 3.5-1, the displacements for the

Transverse Wind load case are amplified by a factor of 5. The cable sags are exaggerated, and can actually be displayed upward if the sag under one load case is less than that of the *Undeformed Geometry*.

Changing the aspect ratio of a displayed structure

The **Z Scale Factor** allows you to exaggerate the vertical coordinates of the positions of all the joints relative to their X and Y coordinates. This may be useful to look at the model of several spans in a transmission line in a single screen view (see Example 6 in Section 5).

Displaying member usage or member axial force by load case

In addition to being able to display joint or member labels or numbers, you can display the percent use (member axial force) of each member for any specified load case. Percent uses are strength checks defined in Section 2.6. For example, if you want to display member usage (member forces) for the *Transverse Load* case next to the *Transverse Load* case deformed geometry, you should select *Usage* (*Member force*) under **Member Labels** and pick *Transverse Load* in the **Labels - Load case** pick box, and make sure that *Transverse Load* is one of the selected load cases in the **Load case geometry display** box.

Color coding of members by strength usage or Tension/ Compression

When you first open the **Deformed Geometry** window, all members of a particular deformed shape are color coded according to their percent strength use. The colors and the corresponding *Percent use* thresholds can be changed in the **Color** portion of the **3-d Controls** box. If you do not want to see color coded shapes, simply check *None (Black)*. You can select *Tension/ Compression* to display all members in tension in blue and all members in compression in red.

Display of maximum member usage or group usage considering all load cases

The *Maximum member usage for all load cases* option allows you to display each member maximum use (color and numerical value), considering all load cases. To use it, you should select *Undeformed Geometry* as the load case in both the **Labels** pick box and **Load case geometry display** areas of the menu.

Other options, such as *Wire frame*, *Render*, *Display fictitious members*, *Maximum group usage for all load cases*, and *Group labels* are only used by our **TOWER** and other structure programs. They are not applicable to **SAPS**.

As was discussed in Section 3.2.1, you can use **View/ Distance between <u>joints</u>** to measure the exact distance in 3-dimension between any two joints or cable points of any two deformed shapes. As you move the mouse on the screen, the first point is automatically selected as that which is closest to the

cursor. Its description as well as the load case for which it is displayed are shown at the bottom of the screen. Then click on the left mouse button to lock-in the first point and start tracking the second point with the mouse. The distance between the two points is displayed at the bottom of the screen.

4. BUILDING A MODEL

After becoming familiar with the various assumptions and examples described in this manual, you will be ready to create your own models after clicking on **File/ New**. The model and associated load cases are created by providing input data in the following menus.

1) Select titles and options in the **General/ ...** menus

2) Select the appropriate element data bases under the **Properties/ ..** menus

3) Generate joints and connect the members with the **Geometry/ ..** menus

4) Create the load cases with **Geometry/ Loads** menu

5) Start the analysis by clicking on **Analysis/ Run**.

6) Look at your results as described in Sections 3.4 and 3.5.

4.1 General menus

4.1.1 General data

Fig. 4-1 shows the **General/ General Data** dialog box. Data in that dialog include:

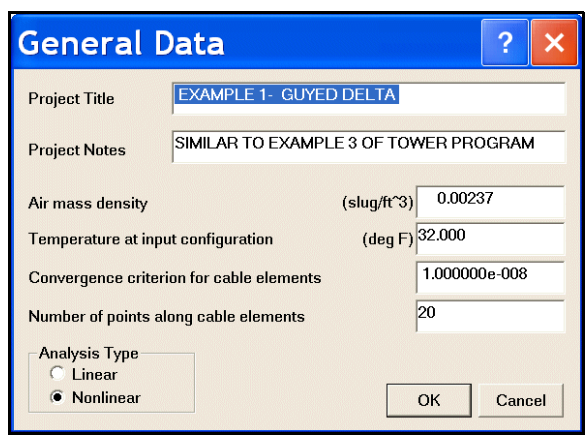


Fig. 4-1 General data dialog

Project title:

Title used on every page of analysis report.

Project notes:

Optional user's notes.

Air mass density:

Mass density of air (or fluid) as needed for the calculation of wind load on elements (see Section 2.4.2).

Temperature at input configuration:

Assumed temperature TEMPIN of all elements for the input locations of the joints. The unstressed length of a truss, beam or fuse element at this temperature is calculated as the distance between the element end joints in the input configuration. If prestress values are input for truss/ fuse elements, they are prestressed at this temperature. In the case of a cable element for which the initial horizontal component of tension is prescribed, the input configuration temperature is used to calculate its unstressed length at the reference temperature of zero degree C.

Convergence criterion for cable elements

The iterative process which determines the equilibrium configuration of a single cable element (Peyrot, 1979) is stopped when the calculated projections of the element are within a certain tolerance of the actual projections. It is necessary that the product of this tolerance by the axial stiffness of the straight cable (AE/L) be much less than the *Maximum Imbalance* specified in the **General/ Nonlinear Convergence Parameters** menu. The tolerance used by the program is the product of *Convergence Criterion* by the length of the cable, L. Therefore, the product *Convergence Criterion x AE* should be smaller than *Maximum Imbalance*. A default value of .00000001 is provided.

Analysis Type:

You can select a linear or a geometrically nonlinear analysis. A linear analysis is acceptable if there are no cable or fuse element and if the joint displacements do not induce significant P-Delta effects. With a geometrically nonlinear analysis, P-Delta effects are accounted for and buckling is detected by either failure to converge to an equilibrium configuration or convergence to a post buckling configuration. A nonlinear analysis should always be used if there are cable elements.

4.1.2 Nonlinear convergence parameters

The **General/ Nonlinear convergence parameters** dialog box is shown in Fig. 4-2. These parameters are only used when you run a nonlinear analysis. The default values should work in most applications.

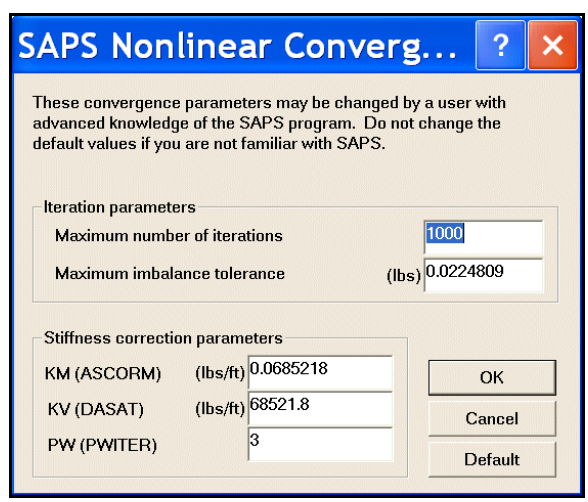


Fig. 4-2 Nonlinear parameters

Maximum number of iterations:

This is the maximum number of times Eq. 2-3 (Section 2.5.2) is applied, unless convergence occurs at a lesser number of iterations.

Maximum imbalance:

This is the maximum force or moment imbalance allowed at any joint for a converged nonlinear solution. Default value in SI units is 0.1 N (or N-m). See

Section 2.5 for the definition of imbalance.

Stiffness correction parameters:

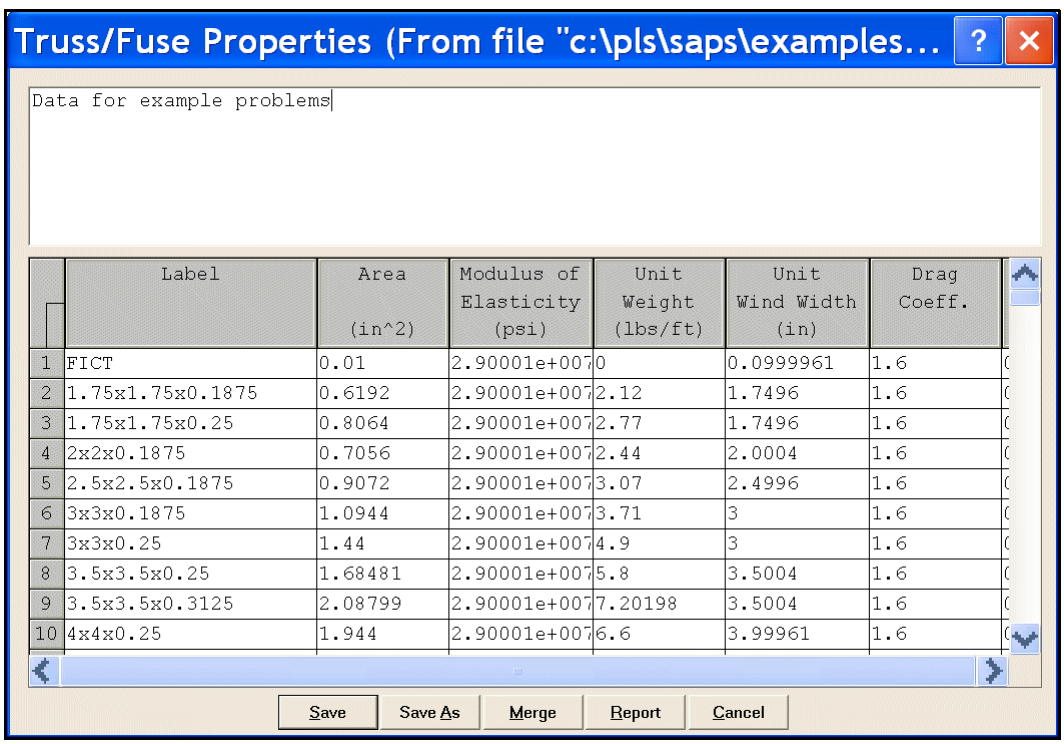
These are the 3 parameters of Eq. 2-2. Default values are KM = 1 N/m, KV = 1,000,000 N/m and PW = 3.

4.2 Element properties data bases

Available elements and their properties are included in files that can be shared by several models. These element property files have the extensions *.tru*, *.bea*, *.cab*, and *.sub* for truss/ fuse, beam, cable and substructure elements, respectively. The data in these files are edited in the four **Properties** tables.

4.2.1 Truss/ Fuse elements

The data for truss or fuse element types (Fig. 4-3) are:



	Label	Area (in^2)	Modulus of Elasticity (psi)	Unit Weight (lbs/ft)	Unit Wind Width (in)	Drag Coeff.
1	FICT	0.01	2.90001e+007	0	0.0999961	1.6
2	1.75x1.75x0.1875	0.6192	2.90001e+007	2.12	1.7496	1.6
3	1.75x1.75x0.25	0.8064	2.90001e+007	2.77	1.7496	1.6
4	2x2x0.1875	0.7056	2.90001e+007	2.44	2.0004	1.6
5	2.5x2.5x0.1875	0.9072	2.90001e+007	3.07	2.4996	1.6
6	3x3x0.1875	1.0944	2.90001e+007	3.71	3	1.6
7	3x3x0.25	1.44	2.90001e+007	4.9	3	1.6
8	3.5x3.5x0.25	1.68481	2.90001e+007	5.8	3.5004	1.6
9	3.5x3.5x0.3125	2.08799	2.90001e+007	7.20198	3.5004	1.6
10	4x4x0.25	1.944	2.90001e+007	6.6	3.99961	1.6

Fig. 4-3 Truss and fuse elements data

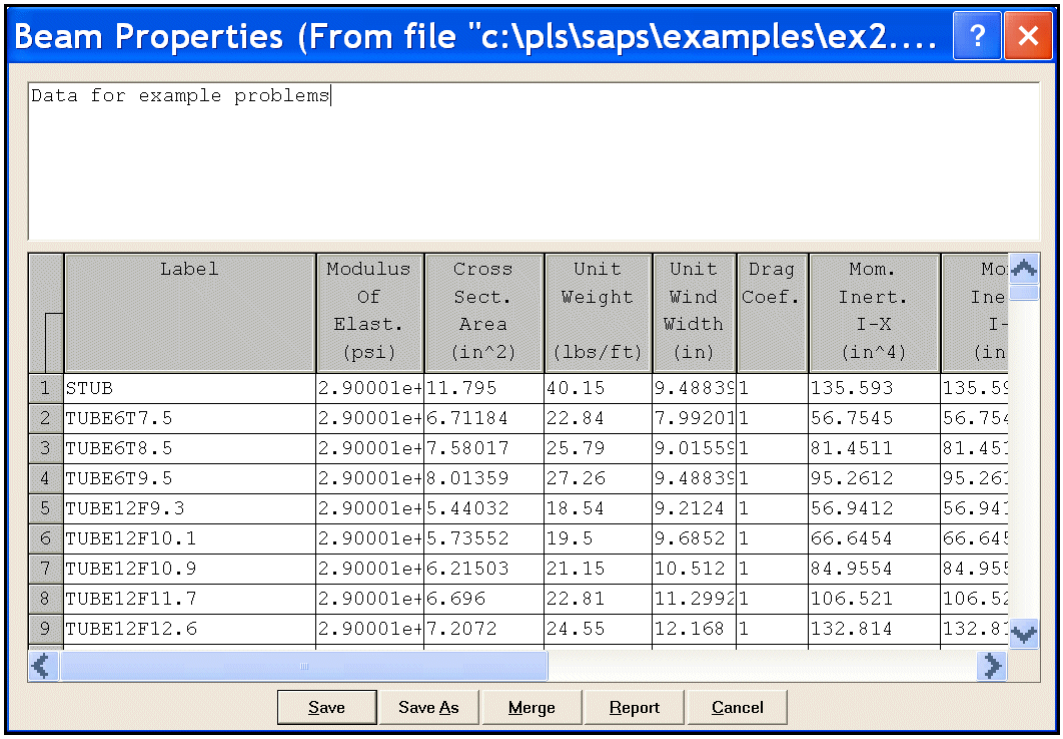
Label: Alphanumeric description of element type (20 characters maximum)

Area, Modulus of elasticity, Unit weight, Thermal expansion coeff.: self explanatory

Unit wind width and *Drag coefficient:* d_e and C_d in Eq. 2-1

4.2.2 Beam elements

The data for beam element types (Fig. 4-4) are:



	Label	Modulus Of Elast. (psi)	Cross Sect. Area (in^2)	Unit Weight (lbs/ft)	Unit Wind Width (in)	Drag Coef.	Mom. Inert. I-X (in^4)	Mo Ine I- (in
1	STUB	2.90001e+	11.795	40.15	9.48839	1	135.593	135.59
2	TUBE6T7.5	2.90001e+	6.71184	22.84	7.99201	1	56.7545	56.754
3	TUBE6T8.5	2.90001e+	7.58017	25.79	9.01559	1	81.4511	81.451
4	TUBE6T9.5	2.90001e+	8.01359	27.26	9.48839	1	95.2612	95.261
5	TUBE12F9.3	2.90001e+	5.44032	18.54	9.2124	1	56.9412	56.941
6	TUBE12F10.1	2.90001e+	5.73552	19.5	9.6852	1	66.6454	66.645
7	TUBE12F10.9	2.90001e+	6.21503	21.15	10.512	1	84.9554	84.955
8	TUBE12F11.7	2.90001e+	6.696	22.81	11.2992	1	106.521	106.52
9	TUBE12F12.6	2.90001e+	7.2072	24.55	12.168	1	132.814	132.81

Fig. 4-4 Beam elements data

Label: alphanumeric description of element type (20 characters maximum)

Modulus of elast., Cross section area, Unit weight, Thermal expansion coef.: self explanatory

Unit wind width and *Drag coefficient:* d_e and C_d in Eq. 2-1

Moments of inertia I-X and I-Z: moments of inertia about local x- and z-axes (see Fig. 2-5)

N.A. dist. C-X and C-Z: distances from neutral axis to extreme fiber for the calculation of bending stress by the classical formula:

(Moment) x (N.A. distance) / (Moment of inertia)

Torsional constant: torsional constant (default used internally is *I-X* + *I-Z*)

Shear mod. elast.: shear modulus of elasticity (default used internally is *Modulus of elasticity* / 2.6)

Shear areas AV-X and AV-Z: shear areas for calculating shear deformations in Planes 1 and 2 (see Fig. 2-7), respectively. Default value used internally is the *Cross section area*.

4.2.3 Cable elements

The data for cable elements are similar to those for truss/ fuse elements (Section 4.2.1).

4.2.4. Substructures

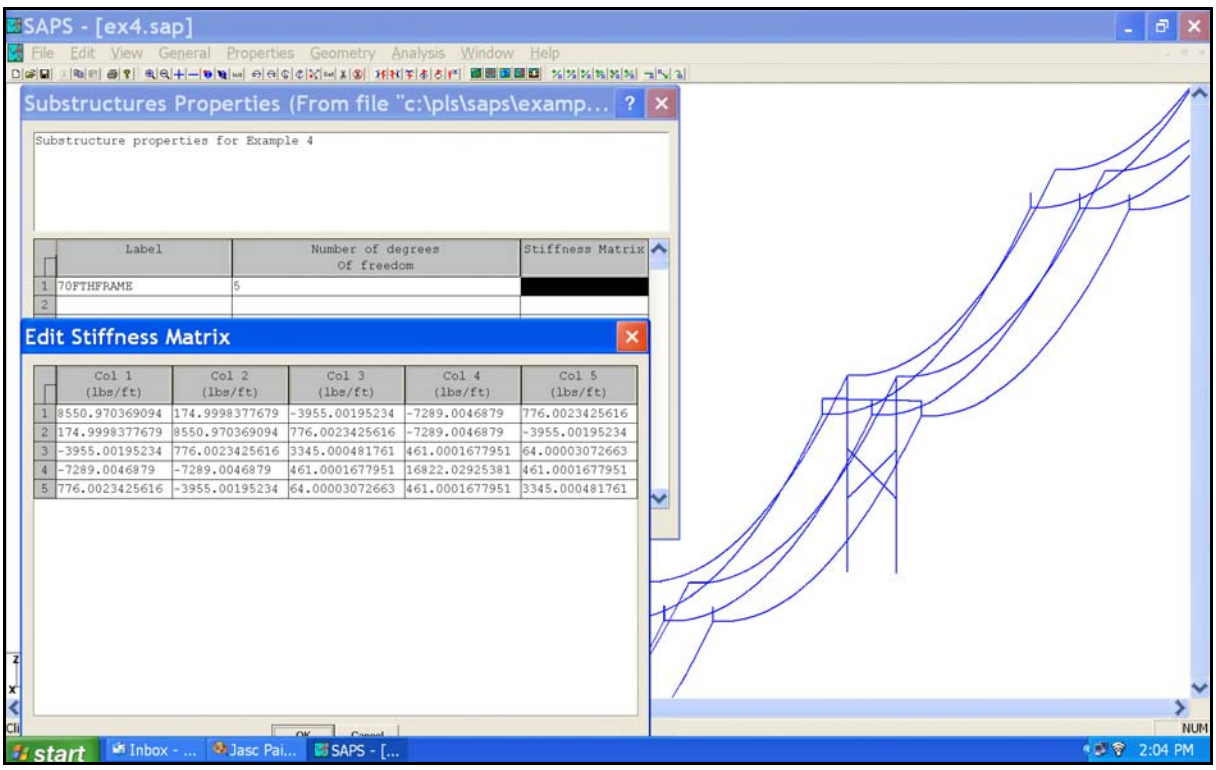


Fig. 4-5 Substructures data

The data for substructures are included in two connected tables. The first table (top part of Fig. 4-5) includes the different substructure types, their labels and total number of degrees of freedom. The second table (lower part of Fig. 4-5) opens when you click on the **Edit Stiffness Matrix** button in the last column of the first table.

The number in *Row i* and *Column j* of the stiffness matrix represents the force needed to keep equilibrium in the direction of degree of freedom "i" when the only nonzero DOF displacement of the substructure is a unit displacement in the direction of degree of freedom "j".

4.3 Geometry

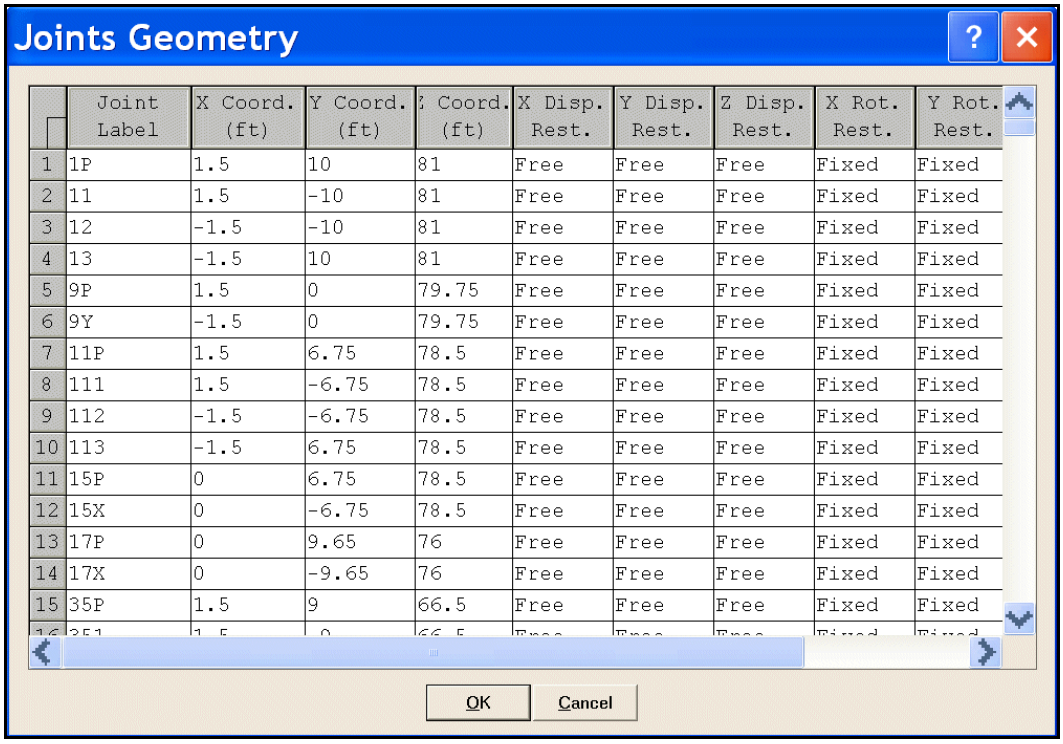
The model geometry is generated by locating joints and installing members between them with the **Geometry/ ..** menus.

4.3.1 Joints

Joints data (Fig. 4-6) include:

Label:

Alphanumeric description of joint (7 characters maximum)



Joints Geometry

	Joint Label	X Coord. (ft)	Y Coord. (ft)	Z Coord. (ft)	X Disp. Rest.	Y Disp. Rest.	Z Disp. Rest.	X Rot. Rest.	Y Rot. Rest.
1	1P	1.5	10	81	Free	Free	Free	Fixed	Fixed
2	11	1.5	-10	81	Free	Free	Free	Fixed	Fixed
3	12	-1.5	-10	81	Free	Free	Free	Fixed	Fixed
4	13	-1.5	10	81	Free	Free	Free	Fixed	Fixed
5	9P	1.5	0	79.75	Free	Free	Free	Fixed	Fixed
6	9Y	-1.5	0	79.75	Free	Free	Free	Fixed	Fixed
7	11P	1.5	6.75	78.5	Free	Free	Free	Fixed	Fixed
8	111	1.5	-6.75	78.5	Free	Free	Free	Fixed	Fixed
9	112	-1.5	-6.75	78.5	Free	Free	Free	Fixed	Fixed
10	113	-1.5	6.75	78.5	Free	Free	Free	Fixed	Fixed
11	15P	0	6.75	78.5	Free	Free	Free	Fixed	Fixed
12	15X	0	-6.75	78.5	Free	Free	Free	Fixed	Fixed
13	17P	0	9.65	76	Free	Free	Free	Fixed	Fixed
14	17X	0	-9.65	76	Free	Free	Free	Fixed	Fixed
15	35P	1.5	9	66.5	Free	Free	Free	Fixed	Fixed

OK Cancel

Fig. 4-6 Joints data

X-, Y-, and Z- coordinates: Joint coordinates in global coordinate system (see Section 2.1)

X-, Y-, and Z- displacements or rotation restraints: Degrees of freedom description:

Free: The joint is free to move in particular direction
Fixed: There is no degree of freedom in that direction, i.e. joint is fixed in that direction
Tied: The joint DOF is tied to that of the master joint (see Section 2.2.2)
Slave: The joint is a slave to the master joint (see Section 2.2.3). In this case, all the displacements and rotations should have *"slave"* selected.

Master joint: Label of master joint. This input is necessary if *"Tied"* or *"Slave"* is selected for any displacement or rotation.

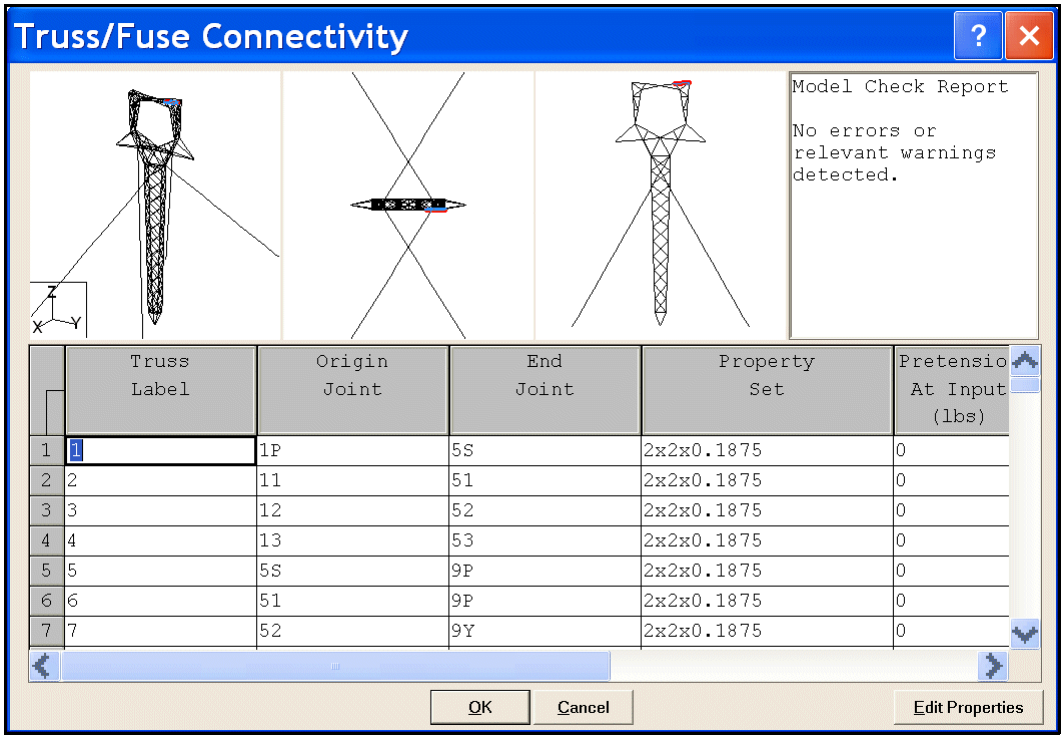
4.3.2 Elements connectivity

4.3.2.1 Truss/ Fuse elements

Data for the truss and fuse elements (Fig. 4-7) include:

Label:

Alphanumeric description of element (7 characters maximum)



	Truss Label	Origin Joint	End Joint	Property Set	Pretensio At Input (lbs)
1	1	1P	5S	2x2x0.1875	0
2	2	11	51	2x2x0.1875	0
3	3	12	52	2x2x0.1875	0
4	4	13	53	2x2x0.1875	0
5	5	5S	9P	2x2x0.1875	0
6	6	51	9P	2x2x0.1875	0
7	7	52	9Y	2x2x0.1875	0

Fig. 4.-7 Truss or fuse connectivity data

Origin and end joints: Labels of joints at origin and end of element. These labels can be picked from the list of available joint labels or can be typed-in followed by *Enter*.

Property set: Element type label as defined in table of Fig. 4-3.

Pretension at input: Force in element (positive if tension) at input temperature defined in the **General/ General data** dialog box for the positions of the joints defined in Fig. 4-6.

Tens. and comp. capacities: Capacities used in Eqs. 2-4 and 2-5 and as absolute values of FCY and FTY as defined in Fig. 2-12 for fuse elements.

Is a fuse?: Indicates whether element is a regular truss element or a fuse. *"Blank"* for truss element, *"Fuse"* for fuse element.
For each separate load case, fuse elements can be inactive (will not be released) or active (will be released if force or deformation exceed specified limits). This is discussed in Section 4.4.6.

ECU/ECY and ETU/ETY: Ratios of deformations defined in Fig. 2-12. Only needed for fuse elements. Set these ratios close to 1, says 1.01 for brittle fuses. For plastic fuses, these ratios can be larger then one. See **Important Note** in Section 2.3.4.

4.3.2.2 Beam elements

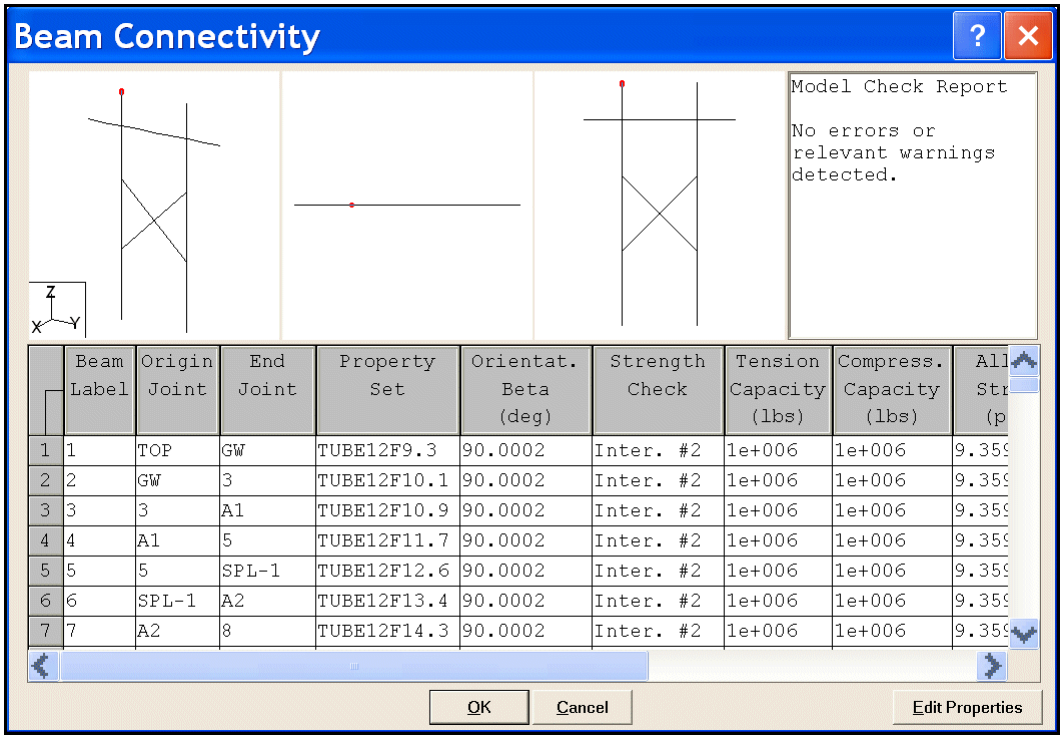
Data for the beam elements (Fig. 4-8) include:

First four columns:

Same as in Section 4.3.2.1.

Orientation beta:

Angle defined in Figs. 2-5 or 2-6



	Beam Label	Origin Joint	End Joint	Property Set	Orientat. Beta (deg)	Strength Check	Tension Capacity (lbs)	Compress. Capacity (lbs)	All Str (p
1	1	TOP	GW	TUBE12F9.3	90.0002	Inter. #2	1e+006	1e+006	9.359
2	2	GW	3	TUBE12F10.1	90.0002	Inter. #2	1e+006	1e+006	9.359
3	3	3	A1	TUBE12F10.9	90.0002	Inter. #2	1e+006	1e+006	9.359
4	4	A1	5	TUBE12F11.7	90.0002	Inter. #2	1e+006	1e+006	9.359
5	5	5	SPL-1	TUBE12F12.6	90.0002	Inter. #2	1e+006	1e+006	9.359
6	6	SPL-1	A2	TUBE12F13.4	90.0002	Inter. #2	1e+006	1e+006	9.359
7	7	A2	8	TUBE12F14.3	90.0002	Inter. #2	1e+006	1e+006	9.359

Fig. 4-8 Beam elements connectivity and other data

Strength check: Type of strength check to be made:

Axial only:	Check of Section 2.6.2.1
Stress #1:	Check of Section 2.6.2.2
Stress #2:	Check of Section 2.6.2.3
Interaction #1:	Check of Section 2.6.2.4
Interaction #2:	Check of Section 2.6.2.5

Tension and compression capacities: Capacities used in Eqs. 2-4 and 2-5 for the *Axial only* strength check.

Allowable stress: Allowable normal stress used in Eqs. 2-6 and 2-7 for the *Stress # 1 and #2* strength checks.

X- and Z-moment capacities: Moment capacities used in Eqs. 2-8 and 2-9 for the

Interaction #1 and #2 strength checks.

Is a fuse?: Indicates whether element is a regular beam element or acts as a fuse axially.
"Blank" for beam element.
"Fuse" for beam element where the axial capacity is limited as that of a fuse. The "fused" beam element acts like a beam in bending but has its axial capacity identical to that of a fuse. This feature does not mimic any real behavior but is used to model tension-only members in our **TOWER** program when the bending stiffness is needed to stabilize a planar joint or a mechanism.
For each separate load case, fuse elements can be inactive (will not be released) or active (will be released if force or deformation exceed specified limits). This is discussed in Section 4.4.6.

ECU/ECY and ETU/ETY: Same as described in Section 4.3.2.1.

4.3.2.3 Cable elements

The data for cable elements are similar to those for truss/ fuse elements (see Section 4.3.2.1) except for the data needed to define the unstressed length of the elements. The unstressed length is computed from the input value of one of the following (**only one value should be entered, not both**):

Horizontal component of tension: Horizontal projection of the average cable tension for the input positions of the end joints at the input temperature in the menu of Fig. 4-1.

or

Unstressed length at zero degree: Unstressed length of cable at the reference temperature of zero degree C.

4.3.2.4 Substructures

Data for each substructure (Fig. 4-9) include:

Substructure label: Alphanumeric description of substructure (20 characters max.)

Property set: Substructure type label as defined in table of Fig. 4-5.

Connectivity: A table with as many rows as there are degrees of freedom for the substructure type (selected property set). Each row lets you define which direction of which joint corresponds to a particular degree of freedom of the substructure. This mapping of each degree of freedom is done by defining:

Global joint label: Label of joint where substructure is connected

Direction: Direction of the degree of freedom:

1 for translation parallel to global X-axis

2 for translation parallel to global Y-axis

3 for translation parallel to global Z-axis

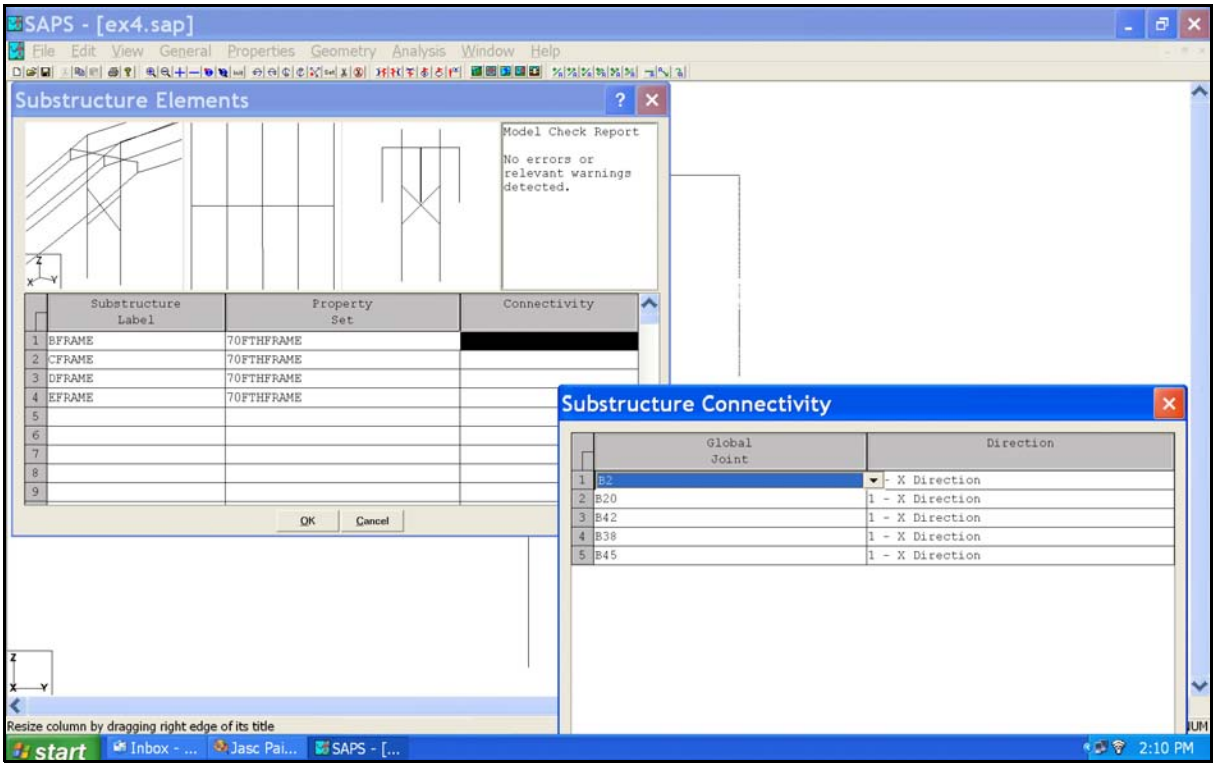


Fig. 4-9 Substructures connectivity

4.4 Load cases

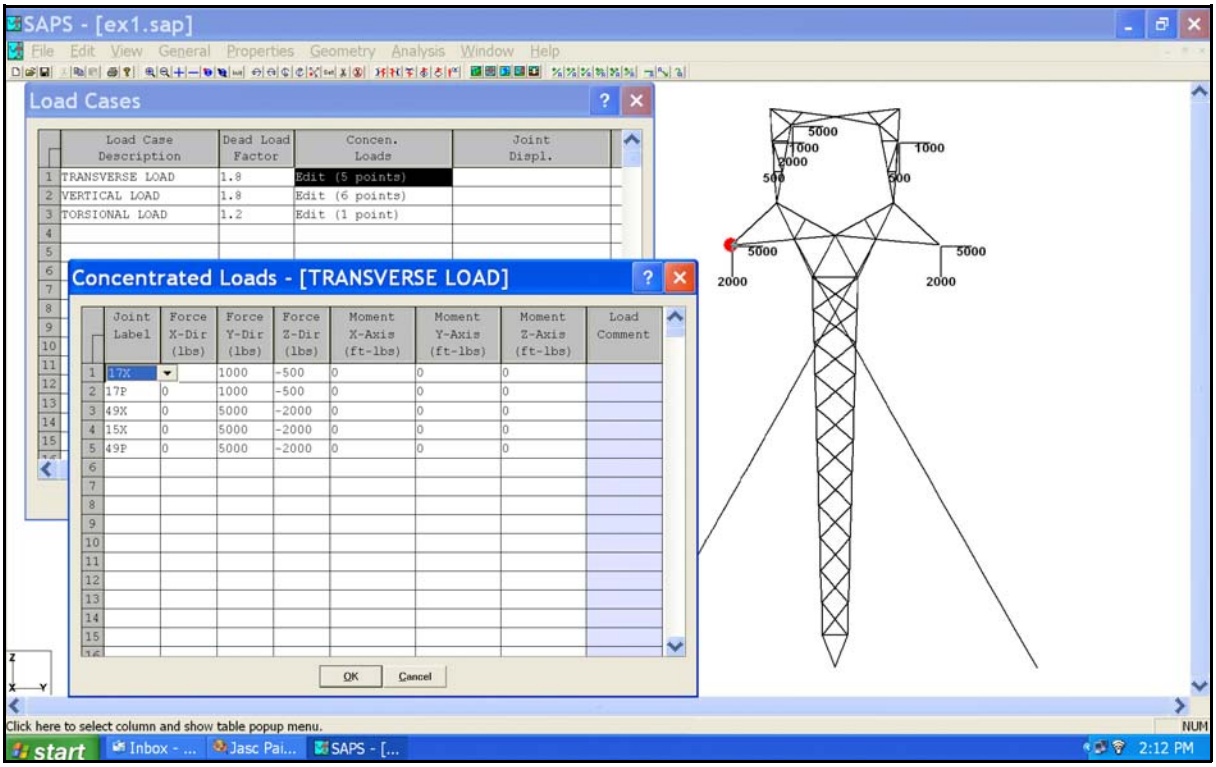


Fig. 4-10 Load cases main menu

Load data are entered in the **Load Cases** table that you reach with **Geometry/ Loads**. Data for each load case (Fig. 4-10) include:

Dead load factor: Factor that multiplies the weight of each element to determine its contribution to the vertical loads on the model. This factor is not applied to the weight of the cable elements as it would invalidate their behavior.

Secondary data defining concentrated loads, support displacements, element temperatures, wind velocity profile, element ice loads and other options. *Secondary data* are included in dialog boxes and tables (see Figs. 4-11 to 4-17) that open when you click on the appropriate column of the main load menu. For example, Fig. 4-10 shows the main load cases menu together with the **Concentrated loads** sub-table that opens when you click in the *Concentrated loads* column.

The secondary data include:

4.4.1 Concentrated loads

Concentrated forces or moments can be applied at any joint in the **Concentrated loads** menu (Fig. 4-10). For each joint with at least one non zero force or moment, the data include:

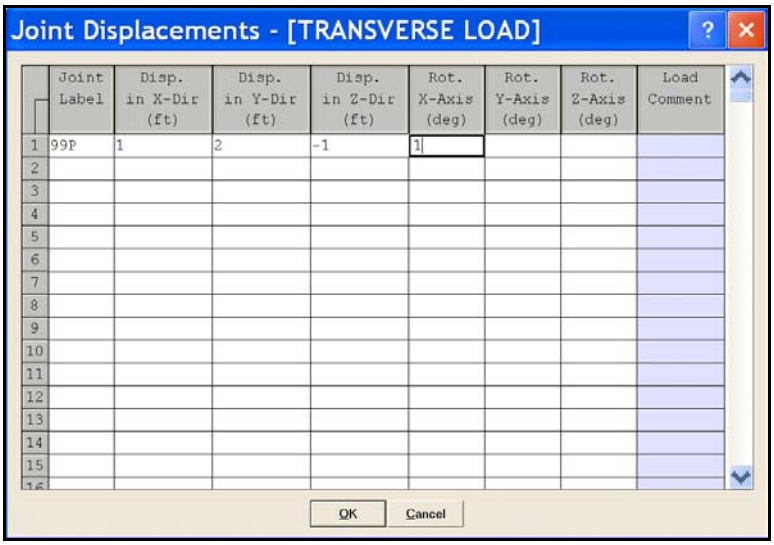
Joint label: Label of joint where force or moment is applied

Force or moment: Numerical value of nonzero component of force or moment in global direction indicated.

4.4.2 Specified support displacements

Displacements or rotations can be specified at support joints. These may represent foundation settlements or rotations. A displacement or rotation can only be specified in a joint direction where there is no degree of freedom, i.e. in a direction defined as *"Fixed"* in the menu of Fig. 4-6.

Data for specified joint displacements or rotation (Fig. 4-11) should be self-explanatory.



	Joint Label	Disp. in X-Dir (ft)	Disp. in Y-Dir (ft)	Disp. in Z-Dir (ft)	Rot. X-Axis (deg)	Rot. Y-Axis (deg)	Rot. Z-Axis (deg)	Load Comment
1	99P	1	2	-1	1			
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								

Fig. 4-11 Specified joint displacements

4.4.3 Element temperatures

Temperatures can be specified globally for all elements or individually for any element. When you click in the *Temperature* column of Fig. 4-10, the **Edit Element Temperatures** dialog box of Fig. 4-12 opens.

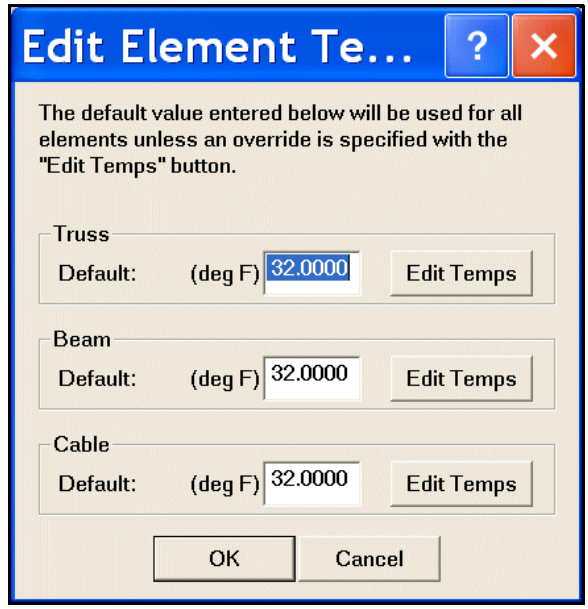
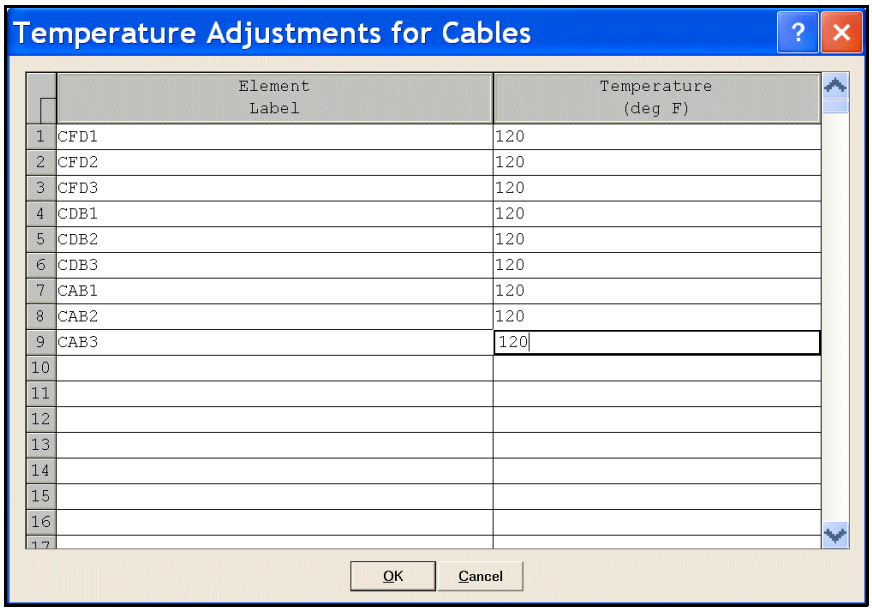


Fig. 4-12 Default temperatures

For the particular load case, the default temperatures (for example 32 deg.F in Fig. 4-12) are applied to all the truss/ fuse, beam and cable elements unless you selectively change the temperature of specific elements by clicking on the *Edit Temps* buttons which open a **Temperature adjustments** table similar to that of Fig. 4-13.

In the **Temperature adjustments** table, you can selectively assign the temperature of any element. For example, in Fig. 4-13, the temperatures of the listed cable elements are specified to be at 120 degrees F. The ability to assign different temperatures to different elements is often needed in models of transmission line spans. For example, one might want to check loads and clearances when some cables are at high temperature (high ampacity conductors) and other are at ambient temperature (ground wires, telephone cables, etc.).



	Element Label	Temperature (deg F)
1	CFD1	120
2	CFD2	120
3	CFD3	120
4	CDB1	120
5	CDB2	120
6	CDB3	120
7	CAB1	120
8	CAB2	120
9	CAB3	120
10		
11		
12		
13		
14		
15		
16		

Fig. 4-13 Specific temperatures

4.4.4 Wind description

For each load case, the wind velocity profile and wind direction are described by the data in the **Wind Profile** dialog box (Fig. 4-14.a) with the optional changes of the **Cable Wind Direction and GRF Adjustments** dialog box (Fig. 4-14.b):

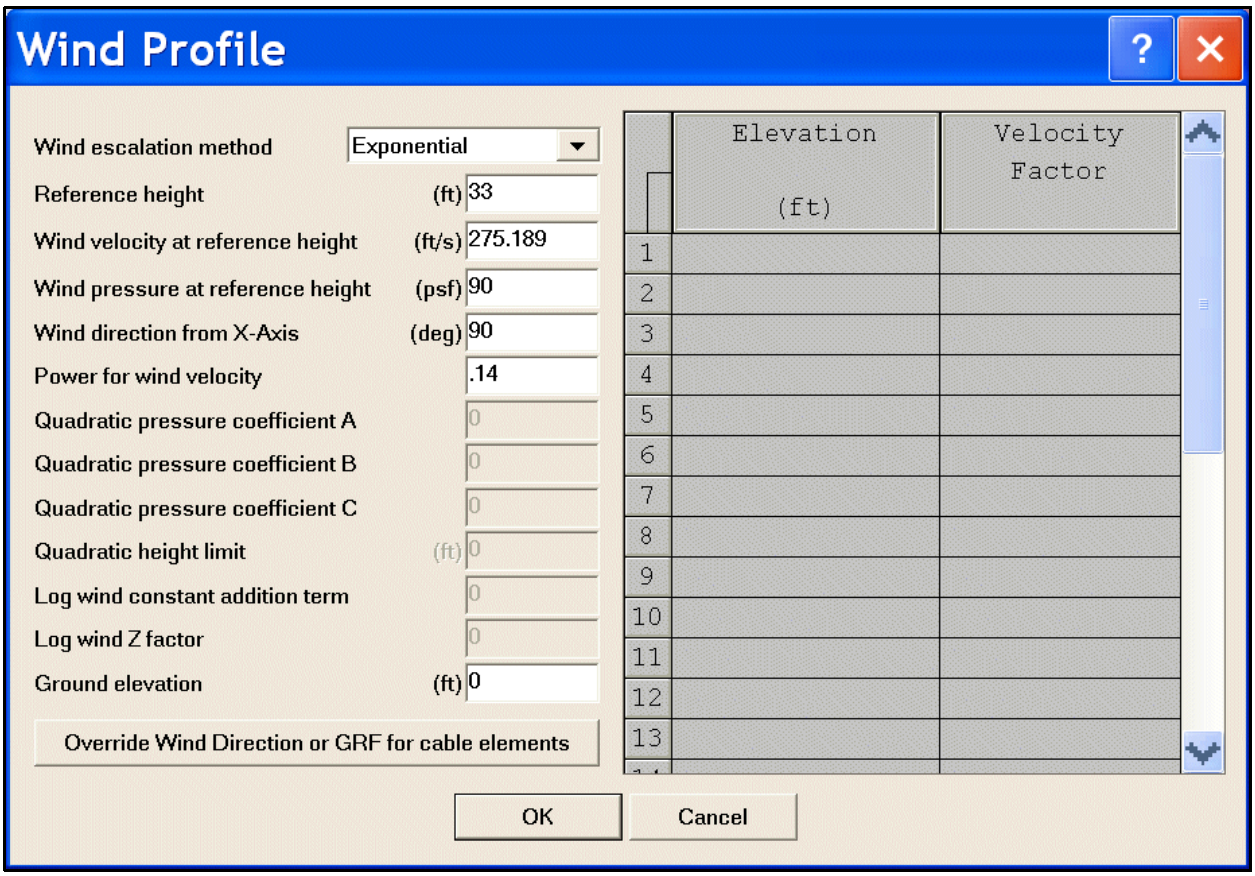


Fig. 4-14.a Primary wind data

Data in the **Wind Profile** box are:

Wind Escalation Method: Method chosen to escalate the wind velocity (V_Z at height Z above the ground) or the wind pressure (P_Z at height Z above the ground):

Constant: The wind velocity at any elevation is equal to that at the reference height, Z_{REF}

Exponential: The wind velocity increase with height " Z " (only valid above the *Reference Height*) is defined by Equation 4-1 (see Fig. 2-14).

$$V_Z = V_{REF} \, (Z / Z_{REF})^{WN} \qquad \text{Eq. 4-1}$$

Logarithmic: The wind velocity increase with height " Z " (only valid above the *Reference Height*) is defined by Equation 4-2.

$$V_Z = V_{REF} (\ln [Z / WN] + E \ Z + F) \qquad \text{Eq. 4-2}$$

Quadratic: The wind pressure increase with height " Z " (only valid above the *Reference Height)* is defined by Equation 4-3.

$$P_Z = P_{REF} (A \ Z^2 + B \ Z + C) \qquad \text{Eq. 4-3}$$

Z in Eq. 4-3 is capped at a maximum value Z_{MAX}.
P_{REF} is defined in Eq. 4-4.

Stair Case: The wind velocity at any height " Z " is equal to the wind velocity at the *Reference Height* multiplied by the *Velocity Factor* entered in the rightmost column in Fig. 4-14.a for the next lower *Elevation* entered in the column immediately to the left.

Linear Interpolation: Same as *Stair Case* except that *Velocity Factor* at height " Z " is interpolated linearly between the values entered in the table.

Reference height: Height " Z_{REF} " at which the basic reference wind velocity or pressure is defined and below which the wind velocity and pressure are assumed constant (see Fig. 2-14 for the " *Exponential* " wind).

Wind Velocity at Reference Height or Wind Pressure at Reference Height:

The wind velocity at the reference height is specified by entering either the actual wind velocity (V_{REF} in Fig. 2-14) or the corresponding pressure P_{REF}. You enter one and the other is calculated using the relationship:

$$P_{REF} = .5 \ \rho \ (V_{REF})^2 \qquad \text{Eq. 4-4}$$

where:

ρ = mass density of moving fluid (air or water): *Air Mass Density* in the menu of Fig. 4-1

Wind direction from X-axis: Angle ALPHA in Fig. 2-14

Power for wind velocity: Only used for " *Exponential* " and "*Logarithmic*" winds:
For " *Exponential* " wind, enter the value of WN in Eq. 4-1.
For " *Logarithmic* " wind, enter the **negative** value of WN in Eq. 4-2
Note: The reason for entering the negative value of WN is to keep compatibility of the newer versions of **SAPS** with older versions that identified the " *Logarithmic* " case with the " minus " sign. The absolute value of WN is actually used in Eq. 4-2.

Quadratic pressure coefficients A, B & C:

Only used for " *Quadratic* " wind. Coefficients A, B & C in Eq. 4-3.

Quadratic height limit:

Only used for " *Quadratic* " wind. Capped Z value Z_{MAX} in Eq. 4-3.

Log wind constant additional term and *Log Wind Z Factor*:

Only used for " Logarithmic " wind. Coefficients F and E in Eq. 4-2.

Ground elevation: Elevation of the ground in the same coordinate system as that used for the **SAPS** model. This value is needed to enable **SAPS** to calculate the mid-height elevation " Z " of each element above the ground from the coordinates of the end points of the element in the model.

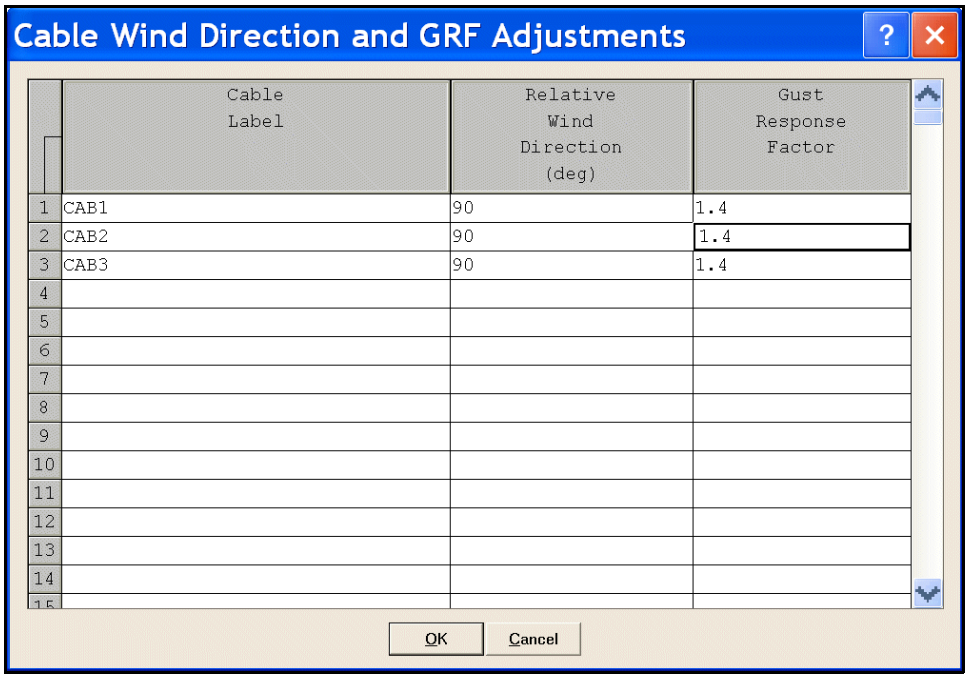
If you click on the *Override Wind Direction or GRF for Cable Elements* button, you will be taken to the table of Fig. 4-14.b where you will be able to override the normal calculation of wind loads on selected cable elements with the alternate procedure described in Section 2.4.2.1. Data in the override cable wind loads include:

Cable Label:

Label of cable for which you want the override calculation (from list of cables in model)

Relative Wind Direction: Angle RWD in Fig. 1-14.a

Gust Response Factor: GRF in Eq. 2-1.a in Section 2.4.2.1



	Cable Label	Relative Wind Direction (deg)	Gust Response Factor
1	CAB1	90	1.4
2	CAB2	90	1.4
3	CAB3	90	1.4
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

Fig. 4-14.b Override cable wind data

4.4.5 Ice loads

Like temperatures, ice loads can be specified globally for all elements or individually for any element. When you click in the *Ice loads* column of Fig. 4-10, the **Ice Accretion on Elements** dialog box of Fig. 4-15 opens.

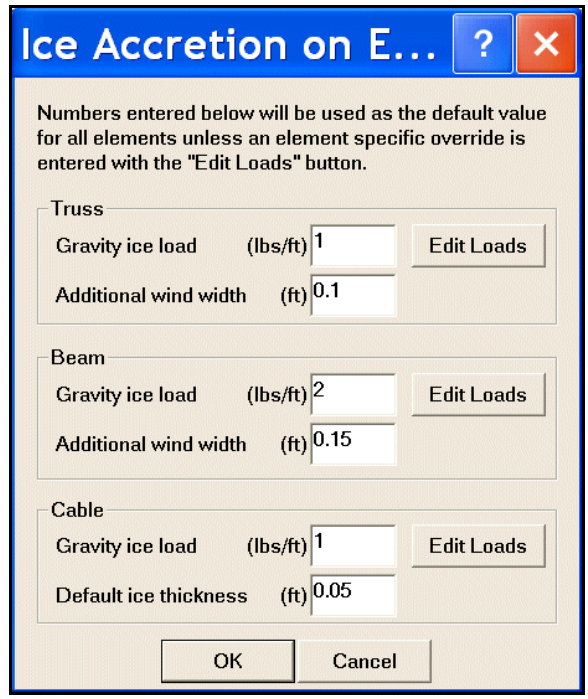


Fig. 4-15 Default ice loads

For the particular load case, the default ice loads and their effect on the wind areas are applied to all the truss/ fuse, beam and cable elements unless you selectively change these values for specific elements by clicking on the *Edit Loads* buttons which open the **Ice accretion for ..** table of Fig. 4-16.

The default data in Fig. 4-15 are:

Gravity ice load:

This is the weight of ice per unit length of element

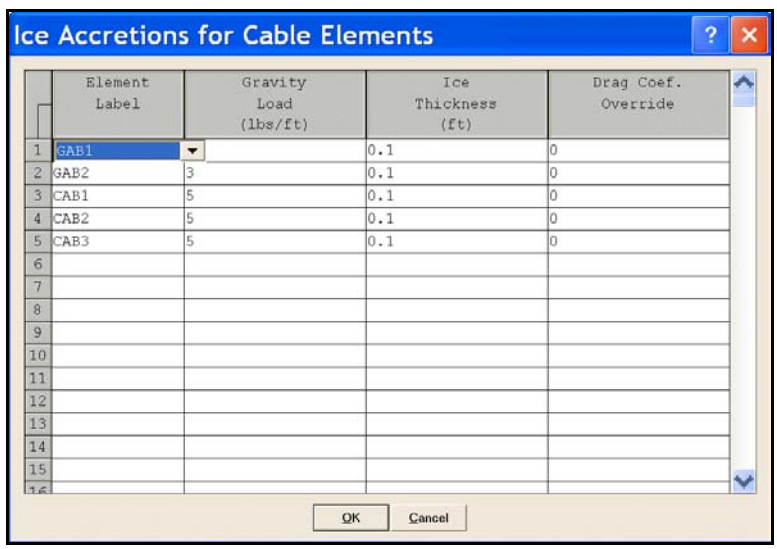
Additional wind width:

For truss, fuse or beam elements, this is the total width of ice that must be added to the element unit wind width to produce the exposed area per unit length, d_e , in Eq. 2-1. This total ice width is generally equal to **twice** the ice thickness.

Default ice thickness:

For cable elements, the ice thickness is input instead of the additional wind width which is calculated internally as **twice** the ice thickness.

In the **Ice accretion for ..** table, you can selectively assign the ice load and additional wind width (or thickness) of any element. For example, in Fig. 4-16, the ice loads of 2 ground wires (GAB1 and GAB2) were set at 3 lbs per foot and those for 3 conductors (CAB1, CAB2 and CAB3) were set at 5 lbs per ft. Ice thicknesses were set at .1 ft.



	Element Label	Gravity Load (lbs/ft)	Ice Thickness (ft)	Drag Coef. Override
1	GAB1		0.1	0
2	GAB2	3	0.1	0
3	CAB1	5	0.1	0
4	CAB2	5	0.1	0
5	CAB3	5	0.1	0
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

Fig. 4-16 Specific ice loads

4.4.6 Load options

The **Load Options** dialog box (Fig. 4-17) lets you select for each load case the following options:

Start with configuration from previous load case:

Lets you implement the strategy described in Section 2.5.3.

Activate after iteration:

By entering a large number, say 1000, the fuse will be inactive. With zero or one, the fuses are activated immediately.

Other items: self explanatory

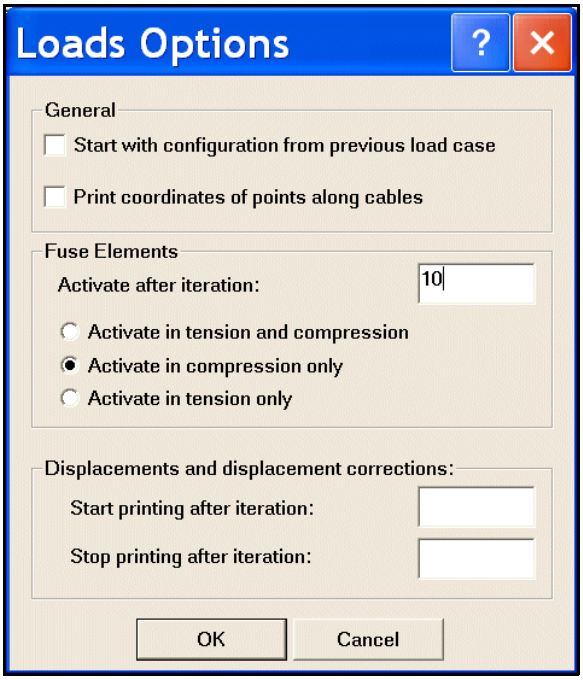


Fig. 4-17 Load options

5. EXAMPLES

The **SAPS** distribution CD contains complete data for the example problems. Therefore, this section provides only brief summaries regarding these examples. The examples use their own element data bases, named *Exi.tru*, *Exi.bea*, *Exi.cab*, *etc.*. However, they could have shared common data bases such as *Shared.tru*, *Shared.bea*, *Shared.cab*. All the geometric and load data for Example "i" are included in the file *Exi.sap*.

Examples 1 and 2 shows how typical nonlinear transmission structures can be modeled.

Examples 3 and 4 are similar, except that in Example 4 some of the frames were replaced by substructures to illustrate the use of substructures. Using substructures to replace actual structures in transmission line models including several structures interconnected by ground wires and conductors substantially reduces modeling and analysis time. This is highly recommended.

Developing the stiffness matrix of the substructures of Example 4 is shown with Example 5.

Example 6 shows how high temperature sags accurately computed with **SAPS** may differ from a conventional analysis with the so-called "ruling span" method.

Example 7 shows how tension-only members often used in latticed tower design can be modeled with fuse elements.

Example 8 shows the modeling of a large complex antenna supported by a guyed mast.

Example 9 shows how to merge existing structures into the current model

5.1 Example 1 - Guyed latticed tower

Features illustrated:

* Truss and cable elements
* Large deformations
* Guyed structure analysis capability

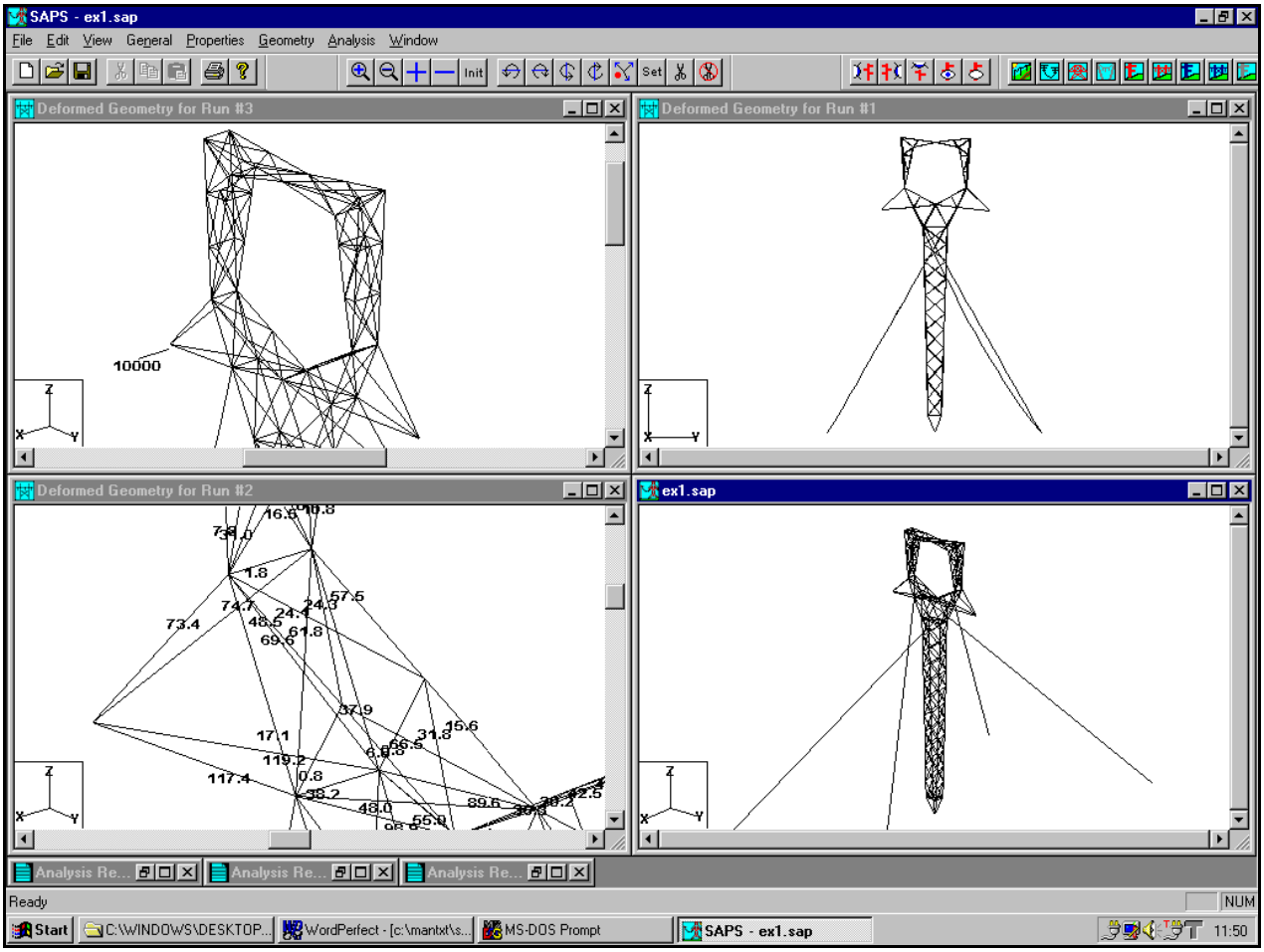


Fig. 5-1 Guyed tower of Example 1

The tower of Example 1 is the same as one of the examples in our **TOWER** program (TOWER, 2006). The undeformed geometry is shown in the lower right corner of Fig. 5-1, the deformed geometry for the broken conductor case in the upper right corner, the broken conductor load in the upper left corner and the percent use of the members in the lower left corner.

You should look at the **Analysis results** report for details of this analysis.

5.2 Example 2 - Steel frame

Features illustrated:

* Truss and beam elements
* Large deformations
* Design checking by various methods

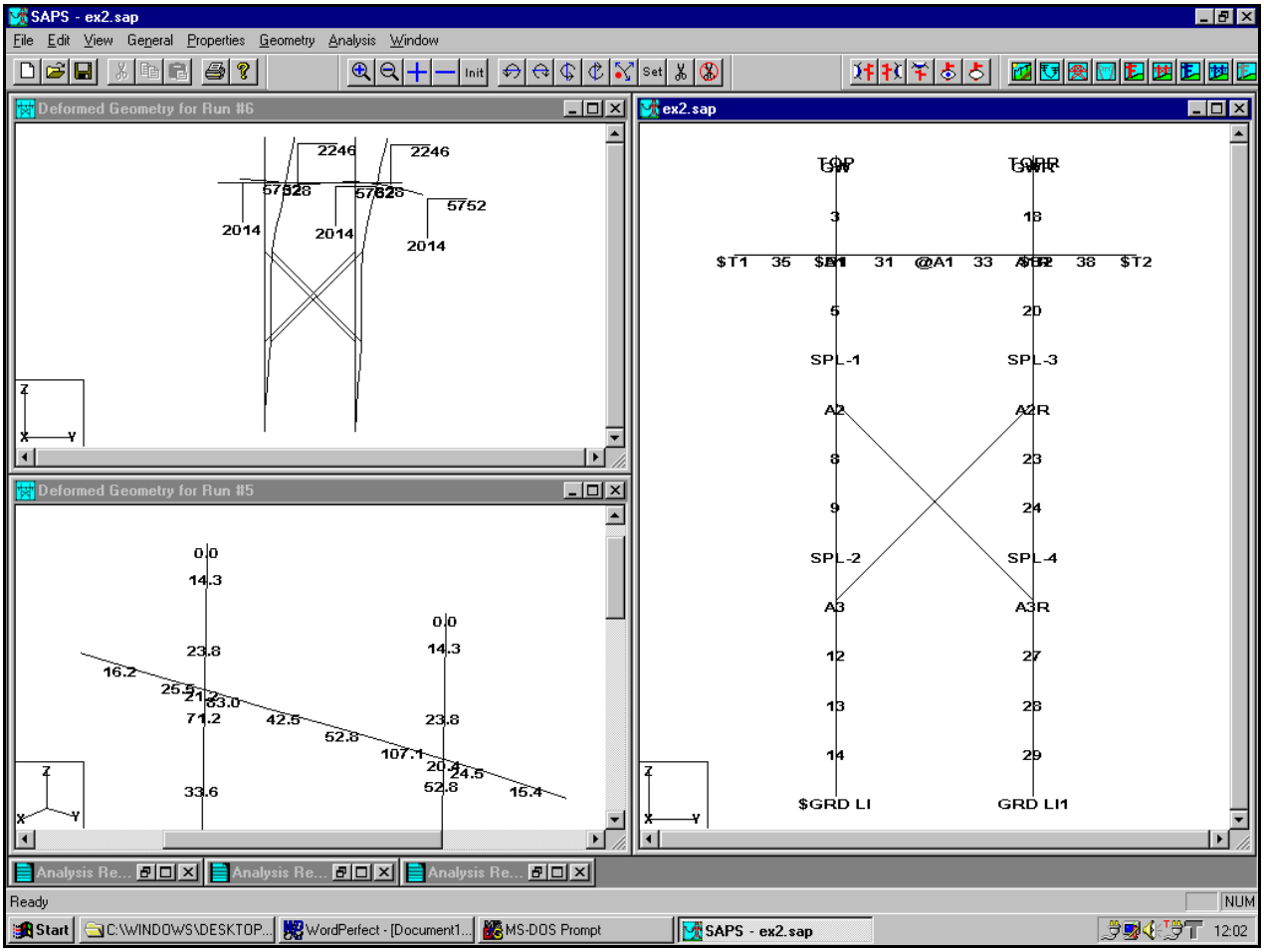


Fig. 5-2 Tubular steel frame of Example 2

The tubular steel frame of Example 2 is similar to Example 3 of ASCE Manual 72 (ASCE, 1990) but with equal arms. The initial geometry is shown in the right portion of Fig. 5-2, the deformed geometry amplified by a factor of 5 for the NESC Heavy load case in the upper left corner, and the corresponding percent member usage in the lower left corner.

5.3 Example 3 - Flexible steel frame line segment (detailed modeling)

Features illustrated:

* Truss, beam and cable elements
* Large deformations
* Unbalanced ice
* Detailed modeling of each individual frame

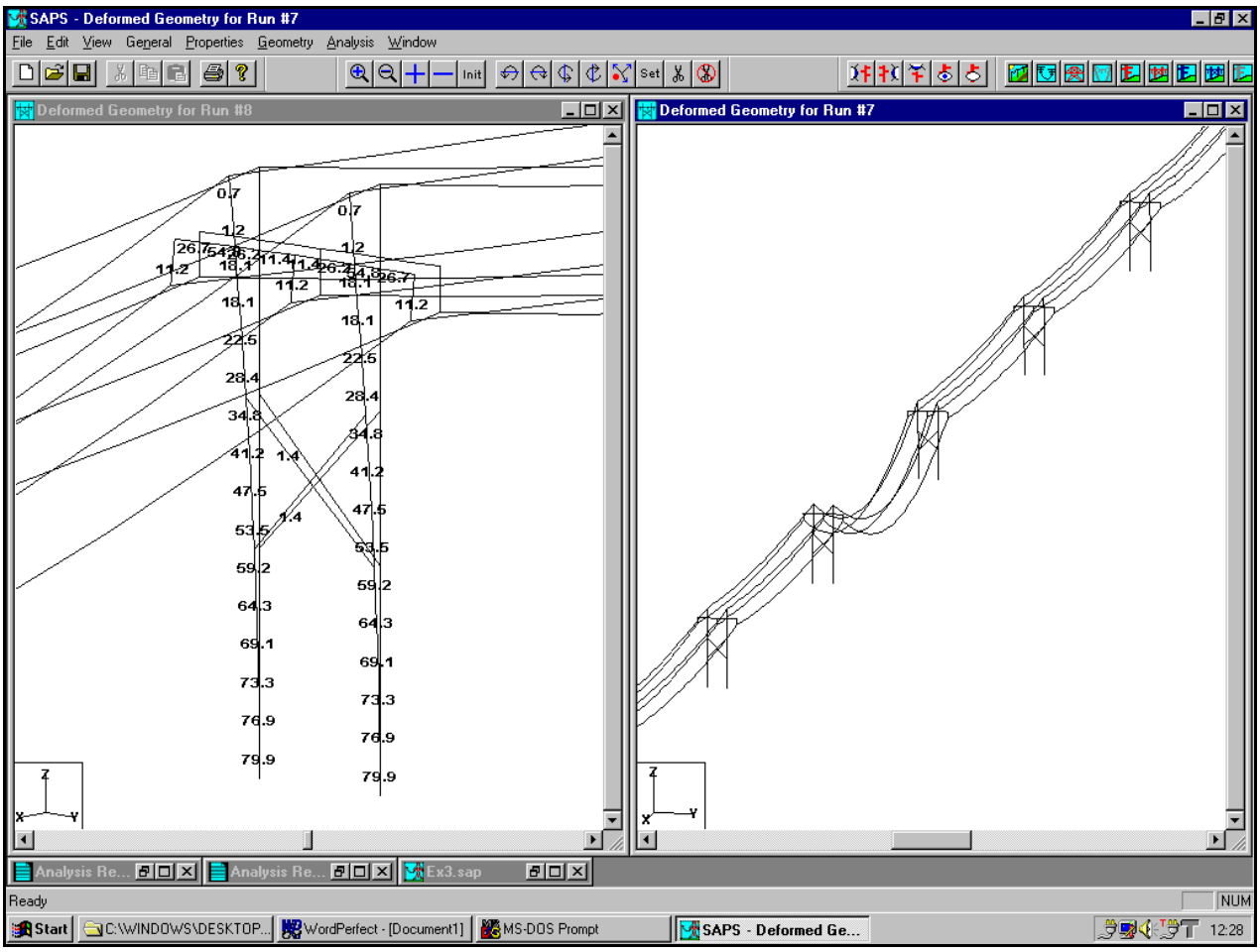


Fig. 5-3 Flexible tubular frames line segment

The line segment of Example 3 consists of five frames supporting six spans. Each frame consists of 40 beam elements to model the poles and arms, 2 truss elements to model the cross braces and three short cables to model the insulators. The model is therefore capable of finding the stresses and displacements in each frame for any combination of weather or other loading event. For example, the right portion of Fig. 5-3 shows the system under heavy unbalanced ice on a single span and the left part shows the deflection (at scale) and usage of the frame next to the span loaded with ice.

5.4 Example 4 - Flexible steel frame line segment (use of substructures)

Features illustrated:

* Truss, beam and cable elements
* Large deformations
* Unbalanced ice, extreme wind, high temperature
* Detailed modeling of one frame
* Modeling of other frames by substructures

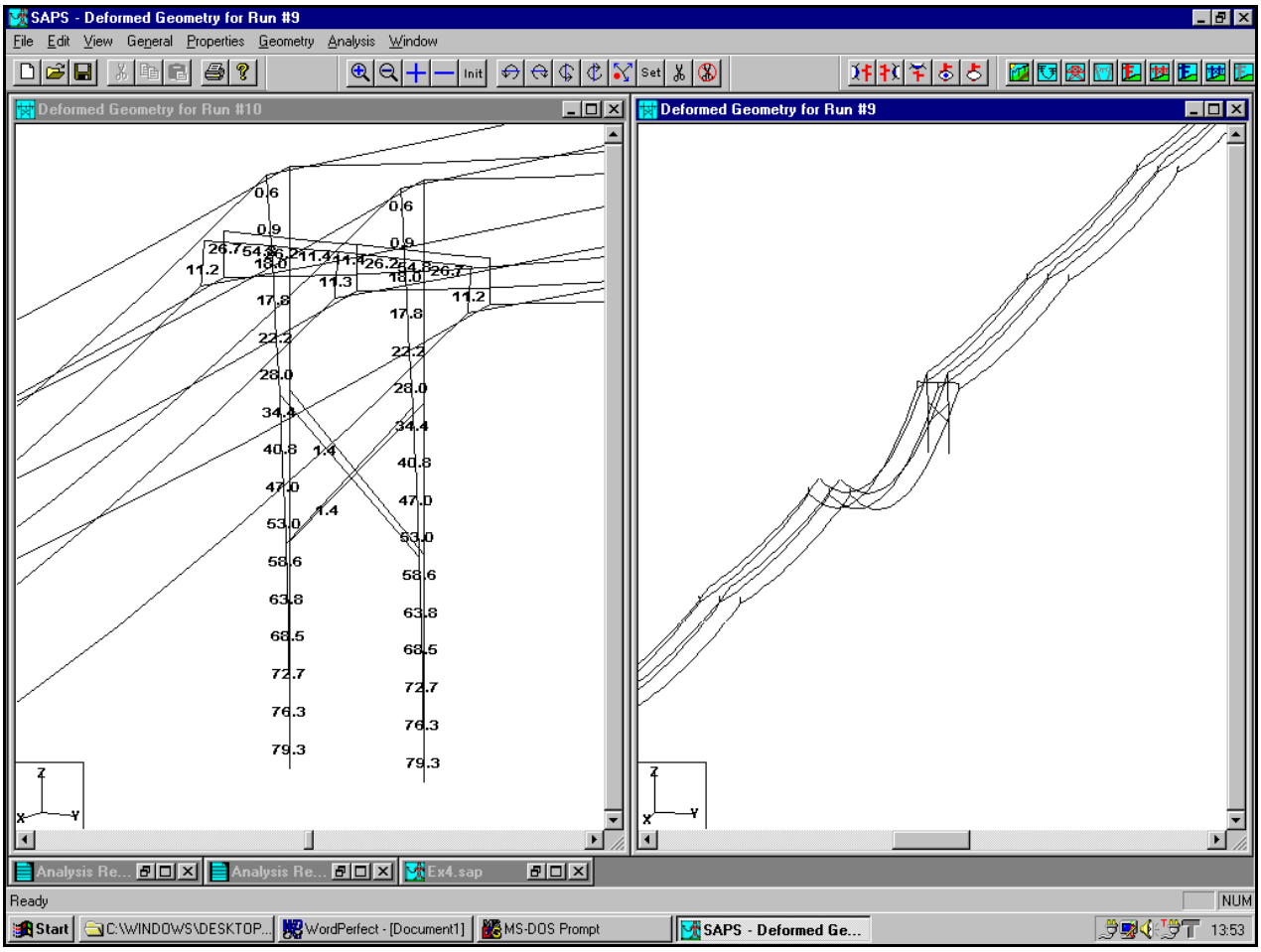


Fig. 5-4 Flexible line segment (use of substructures)

The line segment of Example 4 is identical to that of Example 3, except that four of the frames are replaced by substructure equivalents. The stiffness matrix of the substructure was generated automatically from the basic frame model as shown in Example 5. The small difference between the percent usage at the base of the same frame in Figs. 5-3 and 5-4 (79.3 vs. 79.9) is due to the absence of P-Delta effects on the substructures even though P-Delta effects are correctly accounted for on the detailed frame shown. As can be seen by this example, modeling with substructures can be effective and sufficiently accurate.

5.5 Example 5 - Substructure properties generation

Features illustrated:

* Automatic generation of substructure stiffness matrix

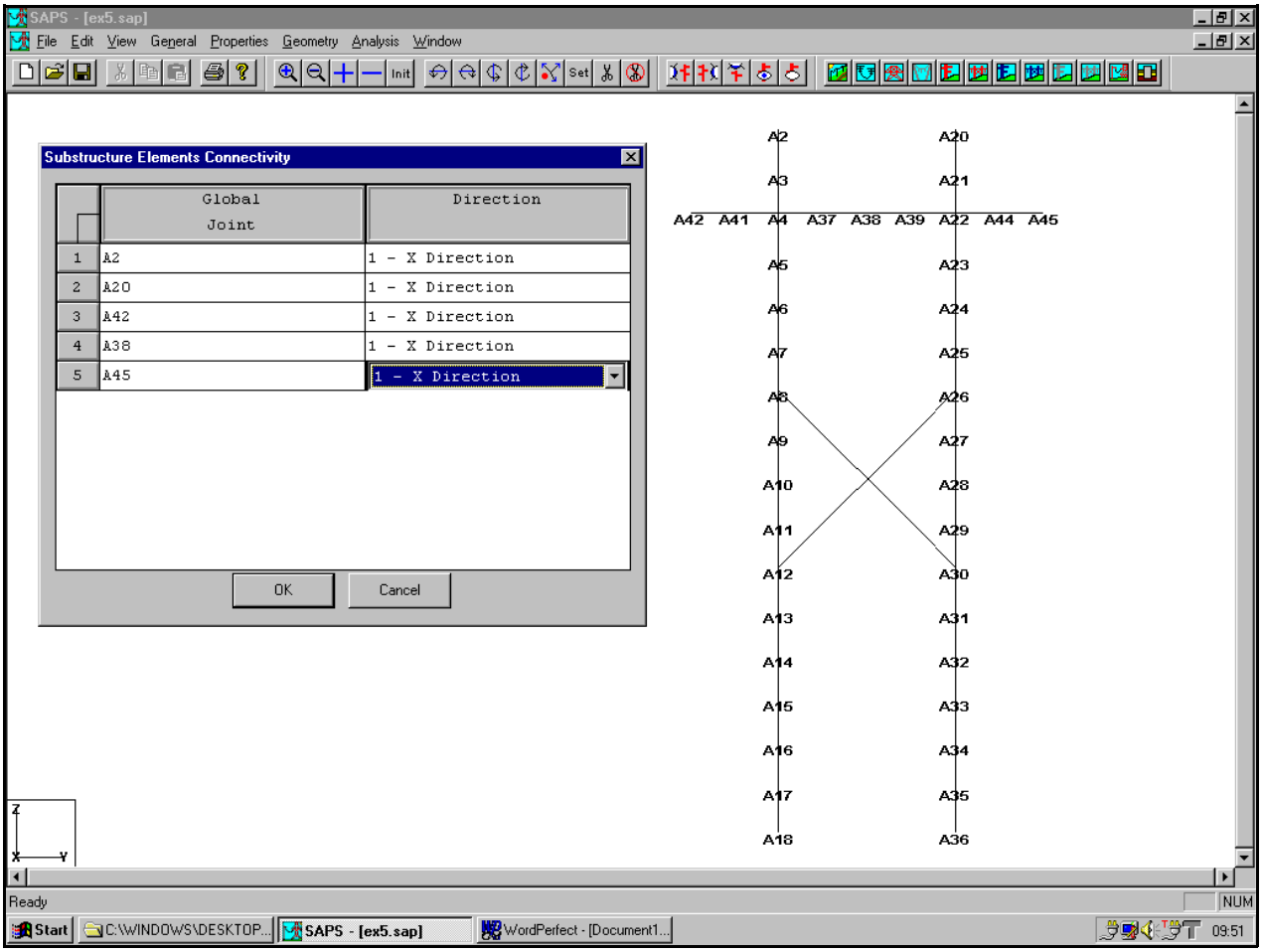


Fig. 5-5 Automatic substructure stiffness matrix generation

This example shows how the stiffness matrix of the substructures used to represent the four H-frames in Example 4 were derived. First you need to have a detailed model of the frame (right side of Fig. 5-5). Then you click on **Analysis/ Create Substructure Model**. You will be asked for the name of the substructure type and the total number of its degrees of freedom (5 in the example). The **Substructure Elements Connectivity** menu at the left of Fig. 5-5 will appear. In this menu you map the degrees of freedom to the joints and directions of the detailed model. When you exit the menu, the substructure stiffness matrix has been created in the *Substructure Elements Properties* file previously named in the **Properties/ Substructures** menu.

5.6 Example 6 - Line segment with different support conditions

Features illustrated:

* Effect of support properties on sags
* Validity of ruling span concept

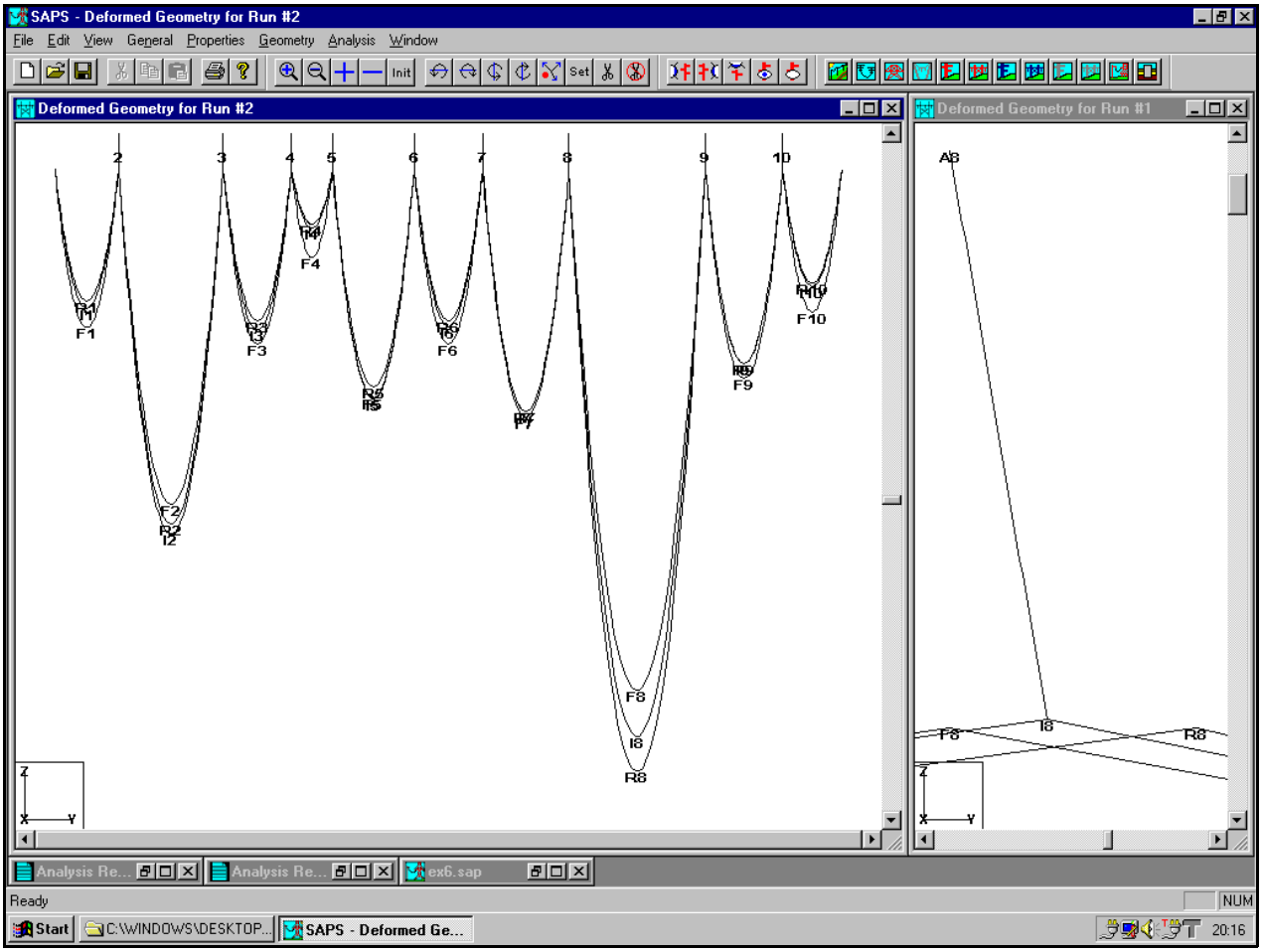


Fig. 5-6 Line segment with different support conditions

The line section of Example 6 is similar to that described in a recent IEEE paper (Motlis et. al., 1998). For the 10 spans labeled I1 to I10, the nine intermediate supports are modeled as 5 ft insulators as in the IEEE paper. For the 10 spans labeled R1 to R10, the supports are modeled as rollers free to move in the Y-direction. For the 10 spans labeled F1 to F10, the supports are fixed. The left portion of Fig. 5-6 shows sag comparisons for the three support conditions at 100 deg. C. The *Z scale factor* of the **3-d Control** box (Fig. 3-4) was set to 80 to get the aspect shown. The right portion of Fig. 5-6 shows, at scale, the displacements of the three supports at the left of the longest span. For the spans supported by rollers, the horizontal component of tension is the same in all the spans, the underlying assumption of the so-called "ruling span" method often used in transmission line sag and tension calculations. The IEEE paper discusses the limitations of the ruling span method.

5.7 Example 7 - Portion of latticed tower

Features illustrated:

* Tension-only members modeled as fuses
* Planar joints and truss mechanisms stabilized with beam elements

Example 7 (Fig. 5-7) includes part of a tower body and associated legs. The diagonal members B1 to B8 as well as the legs internal bracing members D1 to D8 were designed as tension-only members, i.e. members that are supposed to buckle out of the way when loaded in compression in excess of their compression capacity. The diagonals in the K-truss (Members K1 to K8) were designed as tension-compression members. The arrangement of Fig. 5-7 is typical of many older designs in the United States.

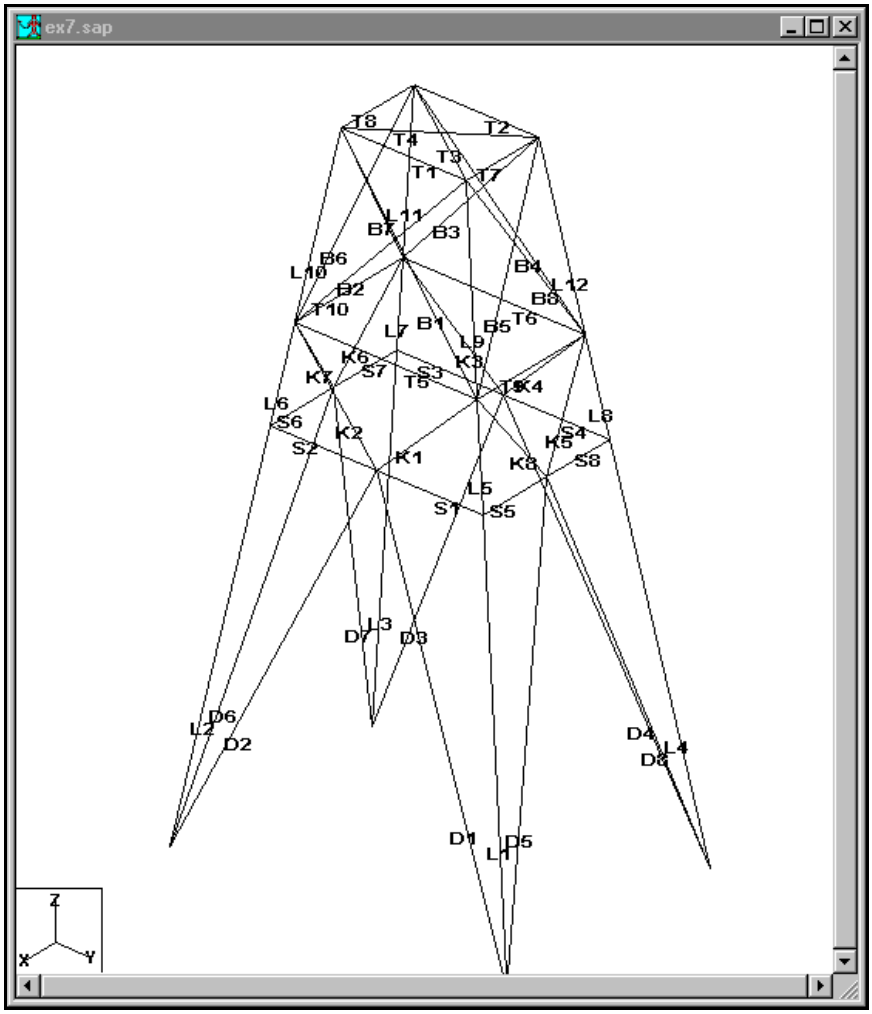


Fig. 5-7 Tower body

In the modeling of Example 7, all tension-only members are modeled as fuses. The corner members L1 to L12 are modeled as continuous beam elements to avoid instability of the tower compression face under lateral load when members B4, B5, D4 and D5 are all removed by the program as tension-only members in compression. The strut members S1 to S8 are also modeled as beam elements to avoid out-of-plane instability of their junction point at the center of the tower faces. Finally, all diagonals or horizontal members that do not need to be modeled as fuses or beams are modeled as truss members.

While **SAPS** can be used to model transmission or communication towers, it is more convenient to use our specialized **TOWER** program for such structures.

5.8 Example 8 - Large antenna

Features illustrated:

* Guyed mast modeled as continuous beam
* Complex antenna system with heavy insulator loads
* Wind and ice on the antenna wires and guys
* Master-slave joints

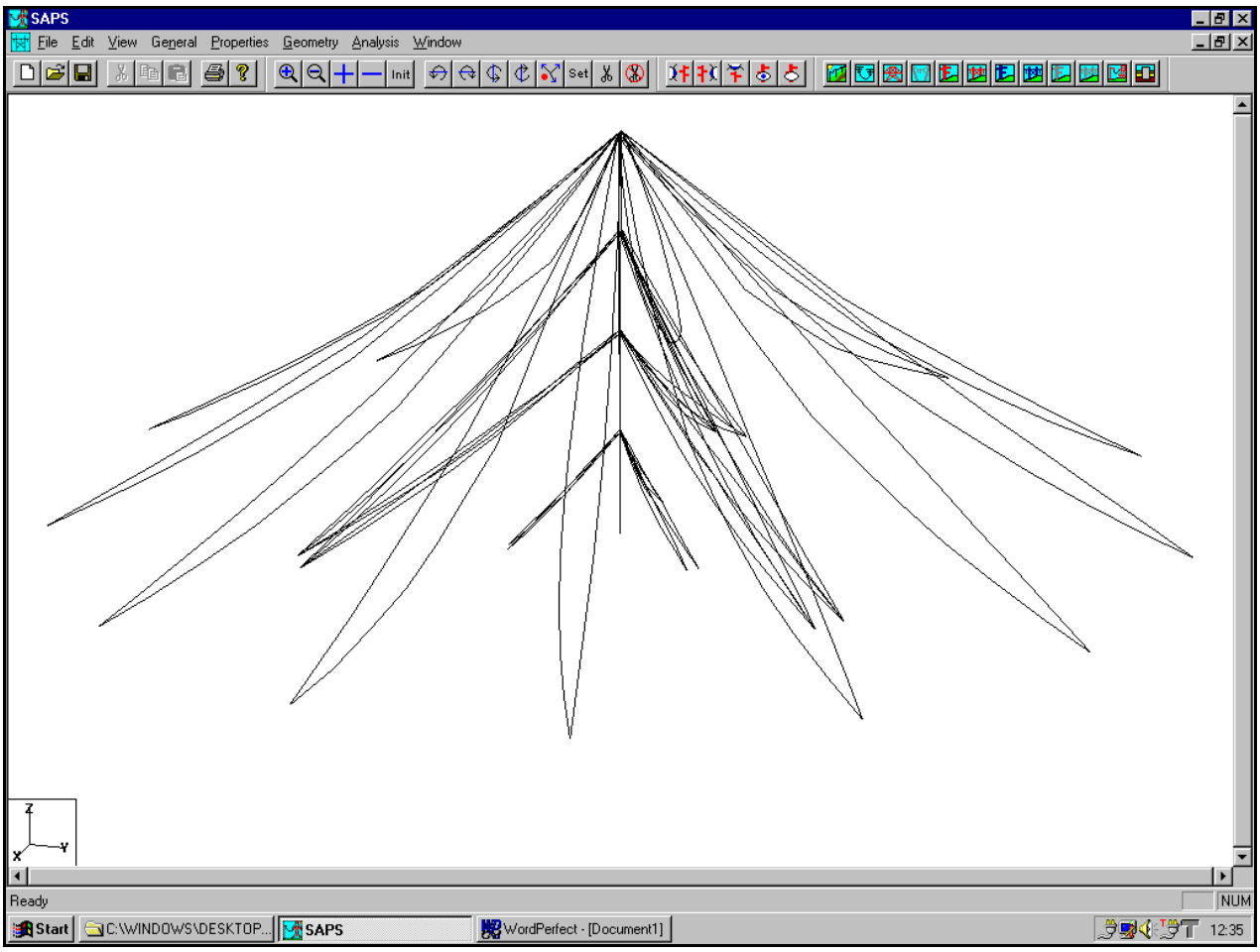


Fig. 5-8 Antenna supported by guyed mast

The guyed mast of the low frequency antenna of Example 8 (Fig. 5-8) was modeled as an equivalent beam. The properties of the equivalent beam were determined from the detailed make-up of the mast which itself was a 3-dimensional truss work. Modeling of a mast by an equivalent beam was popular when computer memory was limited and computer time expensive. However, we now recommend that guyed masts be modeled in greater detail, including all their individual legs, struts and diagonals components. Example 9 of our **TOWER** program (TOWER, 2006) shows such a model. **TOWER** can also determine mast loads by the TIA/ EIA methods and do a complete design check according to those methods.

5.9 Example 9 - Merging models

Feature illustrated: * Merging existing structures into larger model

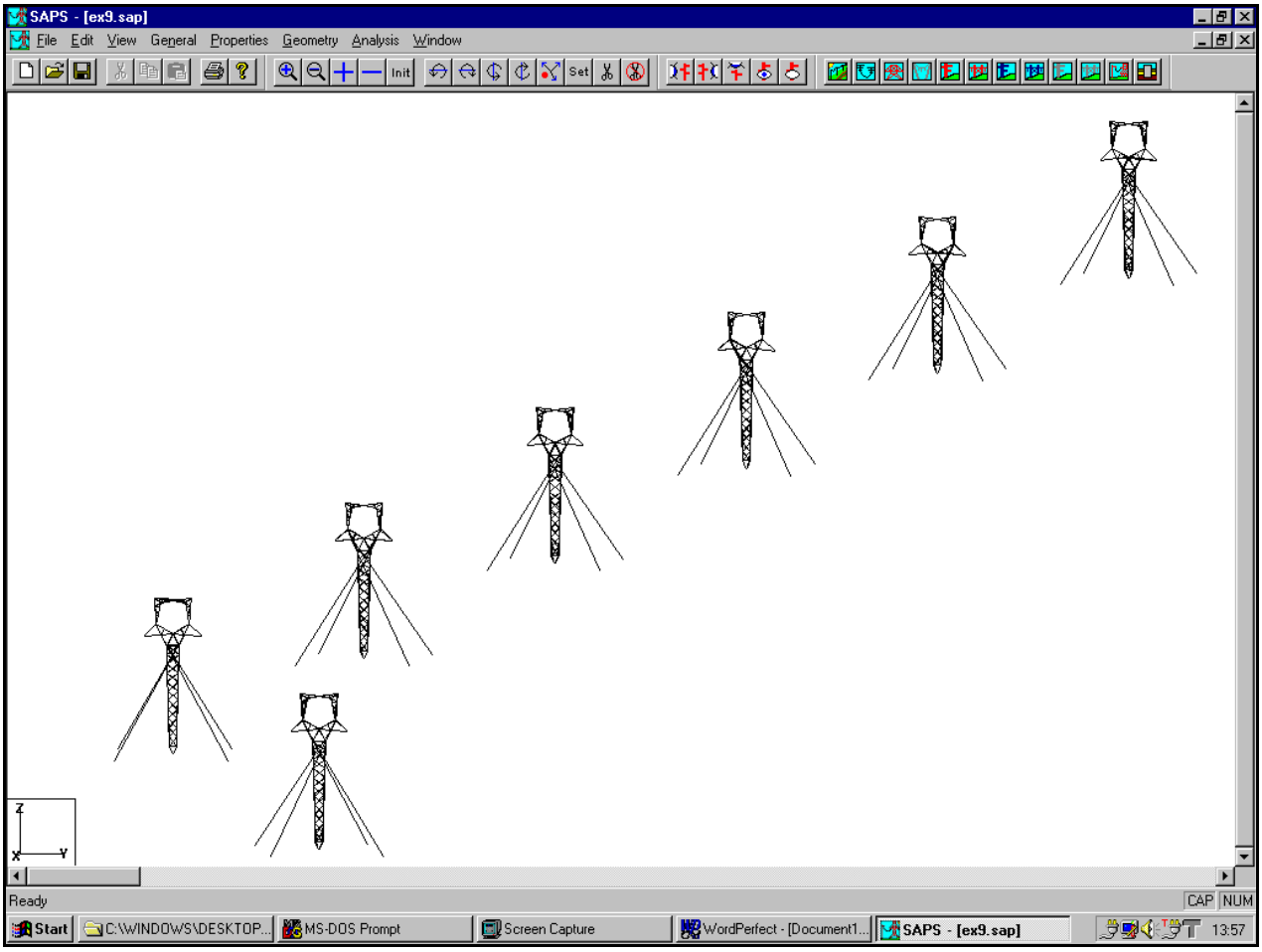


Fig. 5-9 Multi merge of Example 1 tower

Starting with the guyed tower at the upper right of Fig. 5-9, six *Ex1.sap* towers were added to the model by using **File/ Merge model** six different times. Because the leftmost tower in the figure is located at a 10 degrees line angle, it was rotated 5 degrees about the vertical axis when it was merged. The lowest tower in the figure was rotated 10 degrees. Structures other than the *Ex1.sap* tower could have individually been added to the model. As you merge a structure to the model, you are given the opportunity to add a different prefix to the labels of all its joints and members. Make sure that you are using different prefixes to prevent duplicate label names.

This example shows how you can quickly develop the model of a line segment by copying/ pasting individual structure models. The remaining thing to do to finalize the line model is to add span cables (and maybe the insulators). However, a much easier way to create a line model is to use our **PLS-CADD** program which can automatically create a **SAPS** model.

APPENDIX A. HARDWARE REQUIREMENTS AND INSTALLATION

To run **SAPS** you need Microsoft WINDOWS (XP recommended) running on a computer with a minimum of 128 MB of RAM and 40 MB of disk drive space. **SAPS** will take advantage of the 2nd processor in multiprocessor computers.

SAPS is initially shipped on one CD but upgrades are only provided electronically. The program and files can be installed in any directory. The software may be installed on a file server, but the hardware key driver need to be installed on each individual workstation.

UPGRADE INSTALLATION (via E-Mail)

Requesting a program upgrade

Software upgrades are provided exclusively by electronic means. They can be downloaded manually or directly from within the software. In either case you should the **Help/ Download Upgrade** command. This will present the dialog box in Fig. A-1. You will first need to request an upgrade code by clicking on the " *Send Email To Request Code* " button. A code should be E-Mailed to you within one business day. You can then enter the code into the " *I have an upgrade code* " dialog and the program will download the upgrade automatically. Please note that all upgrade codes expire within one to two weeks so you should promptly download your upgrade after receiving the code.

If you should have difficulty downloading the upgrade from within the software, you may download it manually by following the instructions in the upgrade E-Mail. If this case you should use the third option "*I have an upgrade code and upgrade file*" to install the upgrade.

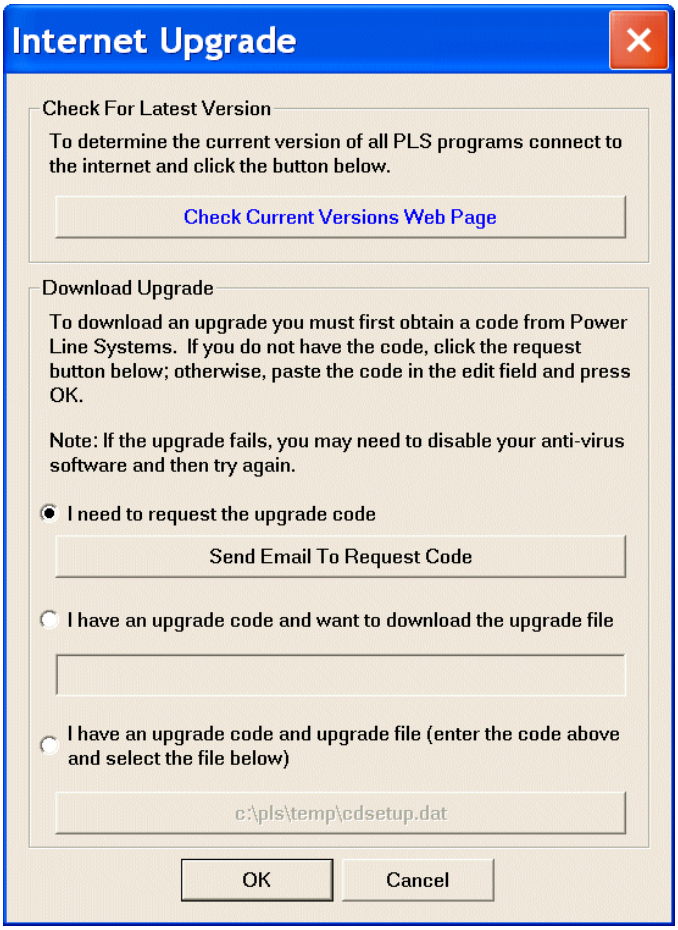


Fig. A-1 Internet upgrade

It may take several minutes before the upgrades you are asking for are downloaded. Once they are downloaded, you will be presented with the same installation dialog box as you do with a **CD Installation** as described later in this section.

Notes for systems administrators and advanced users:

Your upgrade is stored in a sub-directory of the PLS temporary directory named “SETUP”. The PLS temporary directory is defined in **File/ Preferences** and defaults to " C:\PLS\TEMP ". So, if you downloaded an upgrade of **SAPS** it would be stored in “C:\PLS\TEMP\SETUP”. If you need to install the upgrade on multiple computers you may skip the above steps and merely need to copy this directory to the target computer and run the “SETUP.EXE” program. The **Internet Upgrade** feature uses either FTP or HTTP, and will use the proxy settings defined in Internet Explorer. You can verify that you have access to our server by using the **Help/ Check for Updated Manual** command. If this command completes without errors you will know that you can access our servers even before you request the upgrade code. If for some reason you cannot access our servers, you may manually download the upgrade or try modifying your Internet Setting using the “Edit Internet Settings” button in the **Help/ Download Upgrade** dialog.

CD INSTALLATION

During CD installation, you will have the opportunity to install not only **SAPS**, but also the demo versions of all our programs or the commercial versions of all the programs for which you hold valid licenses. The CD installation dialog box (see Fig. A-2) should open after you insert the CD in its drive. If it does not open, click on the **Start** button, then **Run**. When prompted for a file name, select your CD drive and type "*setup.exe*"

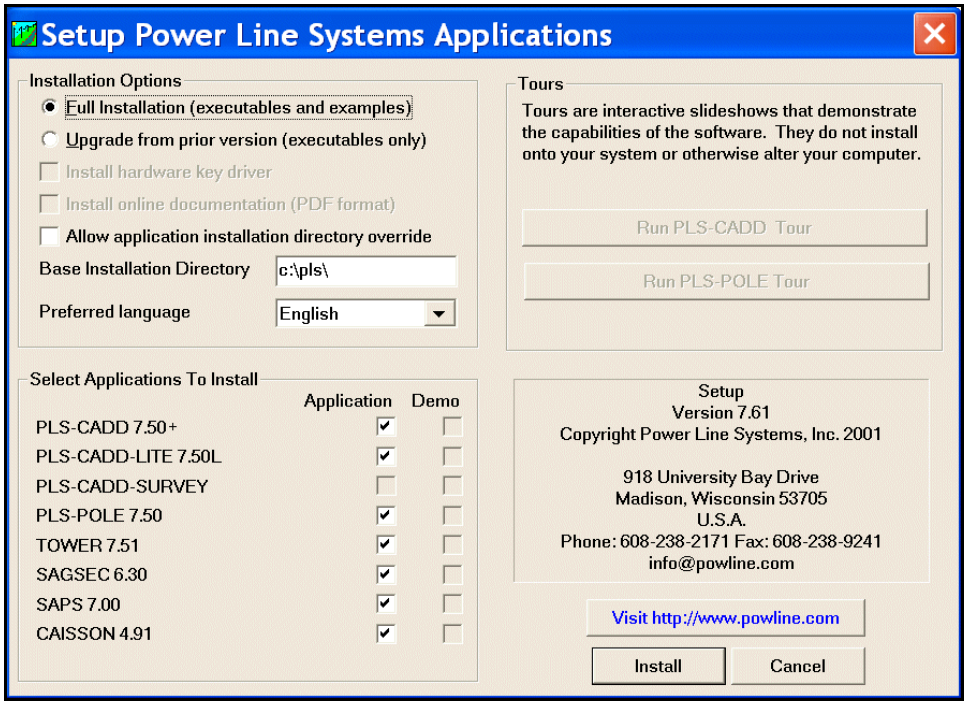


Fig. A-2 Installation dialog from CD

The following options are available when installing from a CD:

Full installation or *Upgrade*: Select the *Full Installation* if the program has not yet been installed or if you want to upgrade the program and it's example files. Select *Upgrade* if you are re-installing over an existing version. You will be prompted " *File Already Exists. OK to overwrite ?* " for each existing file. Respond by clicking on " *Always* ".

Install hardware key driver: This option will check itself automatically as needed. Unless you know that you need or do not need the driver, you should leave this option alone.

Install online documentation (PDF format): This option will check itself if the CD has manuals on it. We strongly recommend that you allow the program to install the electronic manuals. The manuals are in the Adobe (R) Portable Document Format (PDF) which can be viewed using the Adobe Acrobat Reader. If manuals are installed and the Reader is not yet on your system, setup will prompt you after installation to install the Acrobat Reader software. We provide a 32 bit English language version of the Reader. If you want a different language or version you should visit http://www.adobe.com to download it.

Allow application directory override: This option is provided for advanced users to customize the

particular directories that each application will be placed in. It is useful for those who wish to maintain multiple versions of our applications on a computer at once. During the install the program will allow you to type in a directory name other than the default one for each application installed. Unless you feel you have a strong need to do this we suggest you leave it unchecked.

Once you have selected the options above and checked the applications and demos you want installed you may click on **Install** and the program will install all the software.

SPECIAL CONSIDERATIONS FOR NT USERS

If you are using an NT based operating system (such as NT4.0, 2000 or XP), you will need to install the NT driver. To install that driver, check the "*Install hardware key driver*" checkbox in the installation dialog when you install **SAPS**. Windows requires you to have Administrator privileges to install drivers, so if the account you are currently logged into does not have them, you must log out and log in again in an account with Administrator privileges. Follow the instructions on your screen to complete the installation.

TROUBLESHOOTING THE HARDWARE KEY

If your program is displaying a "*Can't Find Hardware Key*" error message, you should consult the following technical note on our web site:

http://www.powline.com/products/ntdriver.html

ELECTRONIC MANUAL AND ONLINE HELP

If you have installed the electronic version of the manual (PDF format) by checking *Install online documentation* in Fig. 0-3, you will have access to online help in any dialog box. All you need to do is to click on the "?" button at the top right of the box and you will be taken to the relevant section of the electronic version of the manual where you will find the appropriate information. Updates to the manual are periodically made available via our Internet site. You should use the **Help/Check for Updated Manual** command to discover if they are available and download them. They are also advertised in our Newsletter which you can read and subscribe to on our web site.

APPENDIX B. REFERENCES

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